Who Lends Matters: Captive Lenders, Deposit Funding, and the Transmission of Monetary Policy

Raffaella Calabrese* Yujia Chen†

Daniel A. Dias‡ Christian Engels§

October 31, 2026

Preliminary and Incomplete

Abstract

This paper combines loan-level data on securitized euro area auto loans from the European DataWarehouse, balance-sheet data on lenders' funding, and high-frequency monetary policy surprises to study how monetary policy shocks transmit to credit terms through lenders that differ in ownership, specialization, and funding. Variation in deposit funding within lender type lets us separate the role of funding from the roles of ownership and specialization. We find that captive lenders raise loan rates and cut loan-to-value ratios, maturities, and loan amounts by more than general banks, also among lenders with similar funding, which gives little evidence that captive lenders insulate borrowers from monetary policy shocks. Specialized lenders keep loan rates in line with those of general banks but, like captive lenders, cut loan-to-value ratios, maturities, and loan amounts. Non-deposit-financed lenders, whose funding costs rise by more after a tightening, set loan rates close to those of deposit-financed lenders but raise loan-to-value ratios, extend maturities, and lend larger amounts, consistent with lenders competing for borrowers through non-price terms.

JEL Codes: E52, G21, G23, L62

Keywords: monetary policy transmission, captive finance, lender heterogeneity, deposit funding, auto loans, credit terms

*University of Edinburgh Business School. Email: raffaella.calabrese@ed.ac.uk.

†Leeds University Business School. Email: Y.Chen9@leeds.ac.uk.

‡Board of Governors of the Federal Reserve System. Email: daniel.dias@frb.gov. The views in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of any other person associated with the Federal Reserve System.

§University of St Andrews Business School. Email: ce50@st-andrews.ac.uk.

1 Introduction

Consumer credit is supplied by banks, by nonbank lenders, and in some markets by lenders owned by the producer of the financed good, and these lenders need not respond to monetary policy in the same way. Banks with less liquid balance sheets or less capital cut lending by more after a monetary tightening (Bernanke and Gertler, 1995; Kashyap and Stein, 2000; Jiménez et al., 2012), and banks with more market power over deposits lose more deposits and contract lending by more (Drechsler et al., 2017). Nonbank lenders, which do not take deposits, respond to policy according to how they are funded (Xiao, 2020; Elliott et al., 2024; Cucic and Gorea, 2026; Buchak et al., 2024; Mimun et al., 2025; Giuzio et al., 2021). Funding is not the only difference across lenders. A captive lender, owned by the manufacturer, earns on the sale of the car as well as on the loan, so it may have reasons that other lenders lack to keep credit terms favorable. Specialized lenders, for which auto credit is the main business, may also respond differently from banks, for which it is one product among many (Benmelech et al., 2017; Jiang et al., 2020). In auto finance, captives lend on different terms from other lenders and respond differently when the parent or the industry is hit (Barron et al., 2008; Murfin and Pratt, 2019; Benmelech et al., 2017; Benetton et al., 2025b; Hankins et al., 2026). Yet studies of funding take the lender's objective as given, and studies of captives rarely account for the fact that some captives take deposits while some banks fund their auto lending with little or no deposits. In this paper we use the case of auto lending in the euro area to study how differences in ownership, specialization, and funding affect the transmission of monetary policy shocks to loan rates and other credit terms.

We focus on three types of auto lenders: general banks (banks hereafter), specialized standalone auto lenders (specialized lenders hereafter), and manufacturer-owned captive lenders. These three lenders are different from each other in terms of their focus – banks have auto loans as one type of loan among many, whereas specialized lenders and captive lenders tend to focus almost exclusively on auto credit (Benmelech et al., 2017; Jiang et al., 2020) – and captive lenders are also different in terms of their ownership, since their lending supports the vehicle sales of the parent manufacturer (Barron et al., 2008; Murfin and Pratt, 2019).[1] But they can also be different with regard to their funding sources – some lenders tend to be more deposit-financed while others rely more on non-deposit funding – and this difference does not line up with the lender type, as several captive lenders hold banking licenses and take deposits, while some bank-owned consumer lenders are funded mainly by their parent bank rather than by deposits, and specialized lenders take almost no deposits

[1] Captive lending is mostly concerned with loans for vehicles of the affiliated manufacturer. The data also contain loans by captives for vehicles of other brands, which we do not treat separately in the main analysis.

and rely on intra-group and wholesale funding. We collect each lender's deposit share from its financial statements, which allows us to compare lenders of the same type with different funding and lenders of different types with similar funding. Each of these differences, in focus, ownership, and funding, is a reason why the same policy shock may reach borrowers differently depending on who lends to them.

A lender's response to a monetary-policy shock depends on how much the shock raises its own marginal cost of funds and on how a loan contributes to its profits. On the funding side, deposit rates tend to adjust slowly to changes in the policy rate whereas non-deposit funding, which is mostly priced at market rates, reprices more quickly (Drechsler et al., 2017, 2021). As such, a lender that is mostly deposit-financed faces a smaller funding-cost shock than a lender that is mostly non-deposit-financed, regardless of whether it is a bank, a specialized lender, or a captive. On the profit side, banks and specialized lenders only make money from the lending, while captive lenders make money from the loan but also contribute directly to the bottom line of the auto manufacturer by helping with sales (Barron et al., 2008). With these key ideas in mind, we combine and adapt existing models of captive finance and of lender competition into a conceptual framework that nests the three types of lenders and both funding sources (Barron et al., 2008; Benetton et al., 2025a). In this framework, each lender's equilibrium response to a policy shock combines the funding-cost shock it faces with the way it and its competitors translate cost shocks into contract terms. By choice, we make the framework flexible enough for the captive's tie to the manufacturer and the specialized lender's expertise in auto credit to affect the transmission of the shock independently of funding. Since lenders set the loan rate, the loan-to-value ratio, and the maturity jointly, a lender facing a higher funding cost does not need to pass it through to the loan rate alone – it may raise the rate and extend the maturity to keep monthly payments from rising as much, or finance a smaller share of the vehicle. Because lenders may respond to the same monetary policy shock by adjusting different margins, we analyze how the three contract terms, and the amount lent, change following a monetary policy shock.

We use three main data sources in our analysis. The first is the European DataWarehouse (EDW), the loan-level repository for public European securitizations, from which we obtain the first reported terms of individual auto loans – the loan rate, the loan-to-value ratio, the maturity, and the loan amount – linked to their origination months, together with vehicle and borrower characteristics and the identity of the originating lender, which we use to classify lenders into banks, specialized lenders, and captives. The European auto ABS market is large, with about EUR 107 billion of public auto ABS outstanding in the fourth quarter of 2024, and captive lenders account for about 70 percent of public securitized auto financing by volume (European DataWarehouse, 2025a,b). The second is lender balance-

sheet data, which we hand-collect from financial statements to quantify each lender's reliance on deposit versus non-deposit funding. Non-deposit funding comprises wholesale funding, such as bonds, securitization, and bank loans, and funding from the lender's parent group, which is generally priced at market rates. The third is the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019), from which we take high-frequency surprises around ECB Governing Council announcements, separating pure monetary-policy shocks from information effects following Jarociński and Karadi (2020). The balance-sheet and shock data are straightforward to use, whereas the EDW data have several peculiarities that require more care – loans enter a securitization pool with a lag that varies across loans and lenders, and the data cover securitized loans only, which are a larger share of lending for captives and specialized lenders than for banks. We restrict the sample to fixed-rate purchase and hire-purchase contracts and distinguish their origination dates from the dates on which their terms are first reported. Section 6 examines whether pool-entry timing and the composition of the securitized sample affect our results.

Our empirical approach is akin to a local projections approach as in Jordà (2005), but in the context of a rich pooled cross-section of loans rather than a panel. The richness of the loan-level data allows us to include a demanding set of fixed effects that control for borrower, vehicle, lender, country, and macroeconomic conditions. Country by origination-month fixed effects absorb the common response of the auto loan market to the shock, so our estimates are differences in the response across lenders rather than the level of pass-through, and fixed effects for the originating lender, the vehicle make, whether the vehicle is new or used, the type of borrower, and the type of loan product absorb differences in the composition of what each lender originates. Taking advantage of the cross-sectional richness of the data, we compare the terms of loans originated within the same country and month by lenders that differ in type, in funding, or in both, and we repeat this comparison at each horizon after a high-frequency monetary-policy surprise, which is by construction unrelated to the conditions lenders face at the time of the announcement. This approach lets us track how the shock reaches the terms of newly originated loans over the following months with one observation per loan.[2] We estimate these differences for the loan rate, the loan-to-value ratio, the maturity, and the loan amount, interacting the shock with lender type, with deposit reliance, and with both together, and we smooth the estimates across horizons following Barnichon and Brownlees (2019) for ease of interpretation.

Our analysis yields three main results. First, captive lenders become more restrictive

[2]Borrower identifiers are not stable across all reporting histories, so the analysis does not track borrowers across separate contracts. A borrower who takes out more than one loan during the sample period therefore appears as separate observations.

than general banks after a monetary tightening, and we find little evidence that captive lenders insulate borrowers from monetary policy shocks. Captive lenders raise loan rates, lower the loan-to-value ratio, shorten maturities, and lend smaller amounts relative to general banks, and the higher loan rates remain when we compare captives and banks with similar funding. Second, specialized lenders are also more restrictive than general banks, but only on non-price terms. The loan rates of specialized lenders move in line with those of general banks, while specialized lenders, like captive lenders, lower the loan-to-value ratio, shorten maturities, and lend smaller amounts. Since specialized lenders share with captives a focus on auto credit but not the ownership tie to a manufacturer, these results suggest that the higher loan rates are specific to captive ownership, whereas the tighter non-price terms are common to lenders focused on auto credit. Third, non-deposit-financed lenders ease non-price terms relative to deposit-financed lenders. Non-deposit-financed lenders set loan rates close to those of deposit-financed lenders but raise the loan-to-value ratio, extend maturities, and lend larger amounts. The same holds within lender type, as captives with less deposit funding raise the loan-to-value ratio and extend maturities by more than captives with more deposit funding. These results are consistent with lenders at a disadvantage in funding costs competing for borrowers through non-price terms. On balance, the results show that lender type and funding both shape how monetary policy shocks are transmitted to credit terms, and that differences across lenders appear as much in non-price terms as in loan rates.

This paper contributes to two main strands of the literature: the literature on the transmission of monetary policy through financial intermediaries and the literature on captive finance. The paper contributes to the literature on monetary policy transmission through financial intermediaries by separating the role of funding from the role of lender type. The bank lending channel literature shows that transmission depends on bank characteristics (Bernanke and Gertler, 1995; Kashyap and Stein, 2000; Jiménez et al., 2012), with the deposit franchise as the main reason bank funding costs move less than market rates (Drechsler et al., 2017, 2021), and a more recent literature compares banks with nonbank lenders, which lack deposits and therefore face a larger funding-cost shock (Xiao, 2020; Cucic and Gorea, 2026; Elliott et al., 2024; Haque et al., 2025; Elliott et al., 2025; Yao and Wang, 2024). In the euro area, pass-through to lending rates has been shown to differ across banks and countries with bank balance sheets and loan characteristics (Altavilla et al., 2020; Vilerts et al., 2025; De Jonghe et al., 2026). In the comparisons between banks and nonbanks, the nonbank is defined by its lack of deposit funding, so differences in funding and differences in the other characteristics of nonbank lenders move together and cannot be told apart. We measure funding directly in a market where it varies within lender type, which allows us to compare lenders of the same type with different funding and lenders of different types with similar

funding. We find that funding and lender type both matter for transmission. Captive lenders raise loan rates by more than general banks also when the two have similar funding, so the difference in transmission between captives and banks does not reflect funding alone, while non-deposit-financed lenders differ from deposit-financed lenders mainly in non-price terms. Comparisons of loan-rate pass-through alone would therefore miss much of how lenders at a funding disadvantage respond.

The paper contributes to the literature on captive finance by studying the transmission of monetary policy through captive lenders relative to non-captive lenders. Existing empirical work on captives has looked at episodes in which the manufacturer or the vehicle is at the center of the shock, such as the collapse of asset-backed funding in 2008 (Benmelech et al., 2017), the Volkswagen emissions scandal (Benetton et al., 2025b), metal tariffs (Hankins et al., 2026), and the pricing of diesel and carbon risk in European auto ABS (Beyene et al., 2026; Fliegel et al., 2025), and has also documented that owning a captive raises the parent's profits and market share (Bodnaruk et al., 2016). In contrast, a monetary policy shock changes the exercise because the shock originates outside the auto industry and raises the cost of funds for every lender in the market. In this setting, we can more directly compare and contrast how captive lenders respond in relation to other types of lenders when faced by the same shock. While the shock is the same, the transmission channels may differ, especially because changes in interest rates affect both the supply of auto lending and the demand for autos. Since the captive is the one lender whose parent loses a sale when higher rates push a buyer out of the market, one would expect captives to have a stronger reason than other lenders to insulate borrowers from the shock, and the more favorable terms captives are known to offer (Brennan et al., 1988; Barron et al., 2008; Murfin and Pratt, 2019) seem to support this view. Our conceptual framework shows that the generosity of captive terms and their response to a shock are separate questions. The manufacturing margin makes captive terms more generous in levels, but its effect on how those terms respond to a funding shock runs only through the captive's market position and through the share of its marginal borrowers who would otherwise not buy, and neither channel pins down the sign or the size of the difference relative to other lenders. Whether captives transmit shocks differently, and in which direction, is therefore an empirical question that the evidence on levels does not answer. We find little evidence that captive lenders cushion borrowers from monetary policy shocks. On the contrary, our results show that captive lenders become more restrictive than other lenders, raising loan rates by more than general banks and specialized lenders, also among lenders with similar funding, and tightening non-price terms.

The rest of the paper is organized as follows. Section 2 presents the conceptual framework. Section 3 describes the data, the lender classification, the funding measures, and the

monetary policy shocks. Section 4 presents the empirical methodology. Section 5 reports the main results, Section 6 examines the robustness of the results, and Section 7 concludes.

2 Conceptual Framework

To guide our empirical analysis, we build on existing models of captive finance and monetary-policy transmission in lending markets. The captive-finance literature emphasizes that lenders can differ in both their lending technology and the role that lending plays within the firm. Independent lenders value an auto loan through the profits of the lending business. A captive finance company is affiliated with an automobile manufacturer and can also take into account how financing terms affect vehicle sales and the profits of the parent. Specialized lenders may in turn have an advantage over banks in originating and screening auto loans. These differences provide natural reasons why loan terms, and potentially their response to monetary policy, need not be the same across lender types.

Our setting adds a second dimension that is central to the empirical analysis: lenders within each of these groups differ substantially in how they fund their loans. Because deposit and non-deposit funding costs respond differently to monetary policy, otherwise similar lenders can face different changes in funding costs following the same policy shock. Monetary-policy transmission can therefore differ because lenders face different cost shocks, because they respond differently to a given cost shock, or both. Following Barron et al. (2008), we capture the captive's incentive to internalize manufacturing profits. Following the lending-market approach of Benetton et al. (2025a), policy affects the terms offered by competing lenders through changes in funding costs. We adapt these elements to three lender types and to the three contract terms observed in our data: the loan rate, the loan-to-value ratio (LTV), and maturity. LTV is the amount financed divided by the vehicle price.

2.1 Lenders, borrowers, and loan contracts

Consider consumers who require financing to buy a vehicle of a given manufacturer at price P. A buyer can finance the purchase through a general bank B, an independent specialized lender S, or a captive finance company C affiliated with the manufacturer. The three financing channels are imperfect substitutes. Borrowers care about the financial terms, but can also prefer one channel over another because of existing relationships, convenience, dealer arrangements, or other factors. The alternative in the modeled market is not to make the financed purchase. Cash buyers are not part of the population considered here.

Lender j offers a contract

$$z_j = (r_j, \ell_j, T_j),$$

where r_j is the loan rate, ℓ_j is the LTV, and T_j is maturity. The initial loan balance is $\ell_j P$. Let $s_j(z)$ denote the fraction of potential financed buyers choosing lender j when the vector of contracts is $z = (z_B, z_S, z_C)$, and let $s_0(z)$ denote the fraction who do not make the financed purchase. Thus

$$s_B(z) + s_S(z) + s_C(z) + s_0(z) = 1, \qquad Q(z) = s_B(z) + s_S(z) + s_C(z)$$

is the number of vehicle purchases generated in the modeled market, after normalizing the mass of potential financed buyers to one.

A loan generates revenues but also requires the lender to fund the outstanding balance and exposes it to credit and servicing costs. We summarize these cash flows with two objects. Let $H(z_j) > 0$ measure the expected discounted amount of funding used by the loan over its life. A higher LTV raises the initial amount financed, and a longer maturity can keep balances outstanding for longer; prepayment and default can also change the amount of funding required. Let $R(z_j)$ denote expected discounted lending returns before funding and screening costs. It includes interest and other loan revenues net of expected credit losses and servicing costs. The three contract terms can affect R through interest receipts, repayment, default, recoveries, and the repayment schedule. Expected lending profit per originated loan is

$$\mu_j(z_j; f_j) = R(z_j) - f_j H(z_j) - a_j, \tag{1}$$

where f_j is the lender's effective funding rate and a_j is a screening and origination cost per loan.

We use a_j to capture a simple advantage from specialization. Specialized lenders are assumed to screen and originate auto loans at lower cost than banks,

$$a_S < a_B.$$

We do not impose an ordering on a_C. Captives may share some of the efficiencies of specialized lenders, but affiliation with a manufacturer does not by itself determine their screening or origination costs.

The ownership distinction enters through the objective of the captive. Banks and specialized lenders maximize lending profits. The captive's contract is chosen as part of the

manufacturer's broader profit-maximization problem:

$$\Pi_B = s_B \mu_B, \qquad \Pi_S = s_S \mu_S, \qquad \Pi_C = s_C \mu_C + MQ, \tag{2}$$

where $M \geq 0$ is the manufacturer's profit from an additional vehicle sale.[3] The margin M can differ across vehicle segments. For used vehicles, it captures the manufacturer's indirect gains from supporting resale values and future demand, which are likely to decline with vehicle age.

The manufacturer earns this margin whether the buyer finances through the captive or through an independent lender. The captive therefore values financing that generates an additional vehicle sale, rather than a transfer of a borrower from another lender to the captive.[4]

This distinction can be stated more precisely. Let D_C denote the fraction of the captive's marginal borrowers who would otherwise not make the financed purchase. An additional captive borrower is then worth

$$W_C = \mu_C + MD_C, \tag{3}$$

compared with $W_B = \mu_B$ and $W_S = \mu_S$ for the independent lenders.[5] In effect, the captive behaves like an independent lender whose cost per loan is lower by MD_C. Ownership therefore makes the captive's terms more generous. It changes the captive's response to monetary policy only to the extent that it changes the captive's market position, or that D_C moves after a tightening, for example because tighter terms at competing lenders leave the captive's marginal borrowers with fewer alternatives.

Lenders choose the loan rate, LTV, and maturity simultaneously. A partial-equilibrium Nash equilibrium is a profile of contracts in which each lender maximizes its objective given the contracts offered by the other lenders, and borrower choices determine lender shares and vehicle sales. Appendix A provides one explicit demand system, assumes a regular interior equilibrium, and derives the local comparative statics used below.

[3] If a fraction λ of buyers who do not make the financed purchase would buy the vehicle by other means, M is replaced by $(1-\lambda)M$; nothing else changes.

[4] The term MQ captures one established reason for captive lending to reflect the objectives of the parent firm. Other motives can also matter. Benetton et al. (2025b) document that captive lending responds to the circumstances of the parent manufacturer, while Murfin and Pratt (2019) emphasize incentives related to resale values. We do not model these motives separately.

[5] Under the logit demand system in Appendix A, $D_C = s_0/(1-s_C)$, which rises when competing offers become less attractive. The interpretation of the manufacturing term as a reduction in the captive's per-loan cost holds for any smooth demand system in which contract terms affect demand through a single index; logit is used only for analytical convenience.

2.2 Funding costs and monetary policy

Funding structure is separate from lender type. Banks, specialized lenders, and captives can each rely to different degrees on deposits and non-deposit funding. Let $d_j \in [0,1]$ denote lender j's deposit share. Around a reference policy rate $\bar{\rho}$, write the effective funding cost as

$$\begin{aligned} f_j(\rho) &= \bar{f}_j + \phi_j(\rho - \bar{\rho}), \\ \phi_j &= d_j\phi_D + (1-d_j)\phi_W, \qquad 0 \le \phi_D < \phi_W. \end{aligned} \tag{4}$$

The coefficients ϕ_D and ϕ_W measure how deposit and other funding costs respond to the policy rate. The assumption $\phi_D < \phi_W$ captures the slower repricing of deposits documented in the deposit-pricing literature (Drechsler et al., 2017, 2021). Other funding consists of wholesale and intra-group funding. Intra-group funding is typically priced off market rates through internal transfer pricing, so it reprices with the policy rate much like wholesale funding. Neither deposit shares nor initial funding costs $\bar{f}_j$ are ordered across lender types.

Because deposit rates adjust gradually, the gap between ϕ_D and ϕ_W is largest shortly after a policy change and narrows as deposits reprice. Differences in transmission driven by funding structure should therefore be most pronounced at short horizons.

Equation (4) gives a direct mapping from funding structure to the cost pressure created by monetary policy. The direct cost effect is the change in lending profit at a fixed contract.

$$\frac{\partial \mu_j}{\partial \rho} = -H(z_j)\phi_j < 0. \tag{5}$$

A lender with a larger ϕ_j therefore suffers a larger decline in lending profit from a tightening at the same contract. Because $\phi_D < \phi_W$, greater deposit reliance reduces this direct cost effect. The result does not require banks, specialized lenders, or captives to have a particular funding structure; it uses the variation in deposit reliance within and across lender types.

Once funding costs change, lenders re-optimize all three terms of the contract. They can raise the loan rate, finance a smaller share of the vehicle, change maturity, accept a lower lending margin, or combine these adjustments. Higher funding costs make contracts that require more or longer-lived funding more expensive. Appendix A.4 shows that, holding the attractiveness of a loan offer to borrowers fixed, a higher funding cost weakly reduces the funding exposure of the lender's preferred contract. This result does not require both LTV and maturity to fall. The way the adjustment is divided across the rate, LTV, and maturity depends on borrower demand and on how the contract affects repayment and losses.

Competition adds a further layer because a lender responds not only to its own funding cost but also to the contracts chosen by competitors. Let $z^*(f_B, f_S, f_C)$ denote a local

equilibrium. Then

$$\frac{dz_j^*}{d\rho} = \sum_{k \in \{B,S,C\}} \frac{\partial z_j^*}{\partial f_k} \phi_k. \tag{6}$$

The ϕ_k terms measure the funding-cost shocks generated by monetary policy. The equilibrium derivatives describe how lenders translate those shocks into contract terms while reacting to one another. Lender objectives, screening costs, demand, and competitive position can all affect these derivatives. Funding and ownership are therefore not competing explanations: ownership can affect how a lender responds to the funding-cost shock it faces.

2.3 Implications for the empirical analysis

The first implication concerns lender type. Banks and specialized lenders have the same standalone lending objective, but specialization can lower the cost of originating and screening auto loans. This can change equilibrium terms and the lender's market position, without implying that specialized lenders must respond more or less to monetary policy once funding-cost shocks are held fixed. Captives also internalize effects on the manufacturing parent. A remaining captive differential among lenders with similar funding structures is therefore consistent with an ownership-objective channel, although it need not identify a particular parent-firm motive. Conversely, similar responses among lenders with comparable funding structures would be consistent with ownership affecting the level of loan terms rather than their response to common funding shocks.

The second implication concerns funding. Greater reliance on funding sources that reprice quickly produces a larger direct cost effect after a tightening, while greater deposit reliance attenuates it. Holding the lender's equilibrium response to funding costs fixed, a larger deposit share therefore implies weaker monetary-policy transmission through funding costs. For loan rates, if own-cost pass-through (the equilibrium response of the loan rate to the lender's own funding cost) is positive, greater deposit reliance attenuates the contribution of the lender's own funding shock to the rate response. The model does not impose the same signed response on LTV or maturity because lenders choose all three contract terms jointly.

Funding structure determines the size of the cost shock, while the lender's objective and market position can affect how that shock is transmitted. We therefore begin the empirical analysis with lender type, deposit reliance, and their interaction. If response differences are closely related to deposit reliance and type differences become small among lenders with similar funding structures, the evidence favors funding exposure as the main source of heterogeneity. If type differences remain, funding alone is insufficient. A type-by-funding interaction indicates that the relationship between funding exposure and transmission differs

across lender types.

Finally, we examine the loan rate, LTV, maturity, and the loan amount, which combines LTV with the vehicle price, rather than equating transmission with the response of the loan rate alone. Similar rate responses can be accompanied by different adjustments in the other terms. A smaller rate response by itself cannot distinguish a smaller funding-cost shock from greater adjustment elsewhere in the contract or in the lending margin. Looking at the four outcomes together shows where adjustment occurs without requiring LTV and maturity to move together.

3 Data

3.1 European Data Warehouse (EDW)

We use loan-level records from the European DataWarehouse on securitized auto loans in Germany, Spain, France, Italy and Portugal. The common estimation sample contains 1,831,887 loans originated by 15 lenders between January 2014 and March 2025. We retain the first reported observation for each resolved contract and assign it to its origination month. We study the first-reported loan rate, original loan-to-value ratio, maturity and loan amount on a common sample with these four outcomes available. Loan rates are expressed in percent; the other three outcomes enter in logs. The data also identify the lender, vehicle make, new or used status, borrower employment and customer category, and loan product.

Table 1 reports 835,560 bank loans, 869,680 captive loans and 126,647 specialized-lender loans. Captive loans have a lower mean rate than bank loans (5.51 versus 6.60 percent), lower mean loan-to-value ratios (76.9 versus 80.6 percent) and shorter maturities (53.9 versus 70.9 months). New vehicles account for 87 percent of captive loans and 56 percent of bank loans. These differences describe the sample; they do not measure responses to monetary policy. Italy contributes 878,505 loans, the largest country share, while France and Portugal have no captive loans (Table 2).

We restrict the sample to purchase and hire-purchase contracts, including contracts with a final balloon payment. We exclude leases and separate residual-value claims because their reported rates and balances need not describe comparable customer credit terms. Rates must be explicitly coded as fixed and have no reported nonzero reset interval.[6] The source records the current rate, rather than a separate original rate. The median delay from origination to the first report is 584 days, the 90th percentile is 1,435 days, and 68.5 percent of loans are first

[6]ESMA requires the interest-rate index field to be coded ND5 for fixed-rate arrangements; missing index information alone is insufficient (European Securities and Markets Authority, 2023, Q&A 5.2.9).

reported more than a year after origination. We use the earliest reported rate and exclude contracts with reported restructuring on or before that observation. Interpreting this rate as the contractual rate at origination therefore still requires that no unobserved repricing occurred before the first report. We do not interpret the rate as an annual percentage rate inclusive of fees.

We identify contracts within lender and pool namespaces, reconcile documented lender successions and corroborated cross-pool repeats, and retain the earliest reporting record for each resolved contract before applying these eligibility rules or requiring observed credit terms. We exclude unresolved identity conflicts and use the loan market of the underlying pool rather than the country label of the source file. The records describe loans that enter securitization pools, sometimes years after origination, and exclude loans retained outside those pools; Section 6 examines pool-entry delays. The results therefore concern credit terms among observed securitized originations; they do not measure application rejection rates or aggregate credit supply.

3.2 Lender balance sheet data

3.2.1 Lender classification

We classify lenders by manufacturer affiliation and business focus. Manufacturer-owned or jointly controlled lenders are captives, including institutions with banking licences such as Volkswagen Bank, Mercedes-Benz Bank and Hyundai Capital Bank Europe. Banks without manufacturer affiliation and with lending activities beyond auto credit form the bank category. The specialized category includes auto-focused finance and leasing companies and licensed institutions focused on auto finance. These categories describe ownership and focus separately from the deposit-funding measure. FCA Bank and the German branch of CA Auto Bank are classified as captives for contracts originated through 2 April 2023 and as specialized lenders from 3 April 2023, when Crédit Agricole became the sole owner. The classification uses the contract date rather than assigning the change to the whole calendar year.

3.2.2 Funding information

We hand-collect lender balance-sheet information from financial statements and measure deposit reliance by the ratio of deposits to total funding. Deposits are deposits from customers outside the lender's corporate group, and total funding is the sum of deposits, intra-group funding, and wholesale funding; non-funding liabilities are excluded. Each lender with usable funding observations receives a fixed classification from its median reported ratio over

2014–2024: high deposit reliance if the median is at least 0.25 and low reliance if it is below 0.25. We refer to lenders with low deposit reliance as non-deposit financed and to lenders with high deposit reliance as deposit financed. Non-deposit funding consists of wholesale funding and intra-group funding. Missing deposit observations remain missing; they are not treated as zero. Lenders without usable funding observations remain in the estimation sample with separate unknown-funding shock interactions. The classification summarizes a lender's funding model over the observation window and is not a predetermined measure of exposure before every policy announcement.

Table 3 shows why lender type cannot substitute for measured funding. Three banks have high deposit reliance and four have low reliance; among captives, two have high and three have low reliance, while one has unknown funding. All four specialized lenders fall below the threshold, so the joint specification has no high-deposit specialized-lender comparison. Banks with high deposit reliance are its reference group, which contains three lenders.

3.3 Monetary policy shocks

To measure monetary policy surprises, we use the Euro Area Monetary Policy Event-Study Database (EA-MPD) of Altavilla et al. (2019). For each ECB monetary policy announcement since January 1999, the EA-MPD reports the change in overnight index swap (OIS) rates with maturities from one week to 20 years, in German, French, Italian, and Spanish government bond yields, in stock price indices, and in euro exchange rates over three intraday windows around the announcement. We use the widest, the "monetary event window," which runs from shortly before the press release that states the policy decision to shortly after the press conference that explains it.[7] Since asset prices before the announcement already reflect what markets expect, and the window is short enough that little other news moves euro rates within it, the change in a rate over the window measures the part of the announcement that markets did not anticipate. We measure this surprise with the change in the one-week OIS rate. The loan-origination window from January 2014 to March 2025 contains 94 announcements. The announcement counts in this subsection refer to that window; lagged responses also use shocks from preceding months.

Changes in the OIS rate around an announcement can reflect news about monetary policy or news about the Governing Council's assessment of the economic outlook (Campbell et al., 2012; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020). If an announcement

[7]Each change runs from the median quote in a ten-minute interval before the press release to the median quote in a ten-minute interval after the press conference: from 13:25–13:35 to 15:40–15:50 Central European Time until June 2022, and from 13:55–14:05 to 15:55–16:05 since July 2022. We use the update of the database of 4 November 2025.

leads markets to expect a stronger economy than they did before it, rates rise because markets expect the ECB to raise policy rates as the economy strengthens, not because the stance of policy has tightened. The literature has proposed several ways to separate the two kinds of news (e.g., Miranda-Agrippino and Ricco, 2021; Andrade and Ferroni, 2021). In addressing this issue, we follow the "poor man's" sign-restriction approach suggested by Jarociński and Karadi (2020). The approach rests on the idea that interest rates and equity prices should move in opposite directions in response to monetary policy news but in the same direction in response to news about the economic outlook. We therefore combine movements in the one-week OIS rate and movements in the Euro STOXX 50 index during the event window to determine whether the news at each announcement is about monetary policy or about the economic outlook. Of the 94 announcements in the origination window, the one-week OIS rate changes at 59, and this approach classifies 26 of them as pure monetary policy surprises.

Since our empirical analysis is at a monthly frequency, we aggregate the pure monetary policy surprises from announcement days to calendar months. A common approach in the literature is to sum the surprises within each month, which ignores when in the month each announcement takes place (Kilian, 2024). We follow the approach suggested by Kilian (2024), which recognizes that a surprise early in the month has more time to affect outcomes in that month than a surprise late in the month. We therefore weight each surprise by the number of days left in the month, including the day of the announcement, so that the monetary policy shock in month t is

$$\Delta\rho_t = \sum_{d=1}^{N_t} w_{d,t}\, s_{d,t}, \qquad w_{d,t} = \frac{N_t + 1 - d}{N_t(N_t+1)/2}, \tag{7}$$

where N_t is the number of calendar days in month t and $s_{d,t}$ is the pure monetary policy surprise on day d of month t, which is zero on days without one. The weights decline linearly from the first day of the month to the last and sum to one. Unlike Kilian (2024), we do not carry the remainder of each surprise over to the following month, because a monetary policy shock should not be serially correlated (Ramey, 2016, pp. 108–110).[8] Of the 135 months in the origination window, the shock is nonzero in 26, one for each pure monetary policy surprise. We normalize the monthly weighted surprise by 0.0504733 throughout. The reported responses therefore correspond to the same increase in this index across specifications. Since the weights depend on announcement timing, this normalization does not correspond to a fixed-size announcement surprise.

[8] In our sample, carrying the remainder of each surprise over to the following month would raise the first-order autocorrelation of the monthly shock from −0.03 to 0.25.

Table 1: Credit terms by lender type

	Mean	SD	25th pct.	Median	75th pct.
Panel A: Banks (835,560 loans, 7 lenders)					
Interest rate (%)	6.60	1.95	4.99	6.52	8.01
Loan-to-value ratio (%)	80.6	21.6	65.6	84.0	100.0
Maturity (months)	70.9	24.0	60.0	62.0	84.0
Loan amount (euros)	15,049	7,361	10,094	14,027	18,983
New vehicle (share)	0.56	0.50	0.00	1.00	1.00
Panel B: Captive lenders (869,680 loans, 6 lenders)					
Interest rate (%)	5.51	2.53	4.10	6.45	7.00
Loan-to-value ratio (%)	76.9	22.2	63.4	80.9	95.5
Maturity (months)	53.9	15.0	42.0	48.0	60.0
Loan amount (euros)	15,833	8,930	9,545	13,402	19,858
New vehicle (share)	0.87	0.34	1.00	1.00	1.00
Panel C: Specialized lenders (126,647 loans, 4 lenders)					
Interest rate (%)	7.83	2.57	6.95	7.95	9.29
Loan-to-value ratio (%)	81.4	23.0	64.7	84.9	100.0
Maturity (months)	67.3	22.9	48.0	60.0	84.0
Loan amount (euros)	17,557	12,765	10,609	14,797	20,576
New vehicle (share)	0.37	0.48	0.00	0.00	1.00
Panel D: All loans (1,831,887 loans, 15 lenders)					
Interest rate (%)	6.17	2.39	4.99	6.50	7.71
Loan-to-value ratio (%)	78.9	22.1	64.5	82.4	98.0
Maturity (months)	62.6	21.8	48.0	60.0	72.0
Loan amount (euros)	15,595	8,618	9,869	13,804	19,500
New vehicle (share)	0.69	0.46	0.00	1.00	1.00

Notes: The sample is the estimation sample of equation (8): loans originated from January 2014 to March 2025 in Germany, Spain, France, Italy and Portugal with the loan rate, loan-to-value ratio, maturity and loan amount observed, together with every fixed effect and the surprise at horizons 0 to 12. Loan-to-value ratios below 10 or above 130 percent, maturities above 120 months, and loan amounts of 100 euros or less or above one million euros are set to missing beforehand. FCA Bank and the German branch of CA Auto Bank appear both as captive lenders and as specialized lenders, so the lender counts by type sum to 17 while the sample has 15 lenders.

Table 2: Loans and lenders by country and lender type

	Banks		Captive lenders		Specialized lenders		
	Loans	Lenders	Loans	Lenders	Loans	Lenders	All loans
Germany	48,278	1	156,116	3	8,874	1	213,268
Spain	289,713	2	218,805	1	–	–	508,518
France	98,219	1	–	–	6,027	1	104,246
Italy	287,730	1	494,759	2	96,016	1	878,505
Portugal	111,620	2	–	–	15,730	1	127,350
All	835,560	7	869,680	6	126,647	4	1,831,887

Notes: Counts refer to original-lender LEIs in the securitized-loan estimation sample of Table 1, after applying the data-availability restrictions. Coverage varies by country and lender type. A lender is counted in each country in which it lends, and lenders that change type are counted in each type in which they appear.

Table 3: Lenders by deposit reliance

	High deposits		Low deposits		Unknown	
	Lenders	Loans (%)	Lenders	Loans (%)	Lenders	Loans (%)
Banks	3	26.7	4	73.3	0	0.0
Captive lenders	2	40.8	3	54.3	1	4.9
Specialized lenders	0	0.0	4	100.0	0	0.0

Notes: Each lender is classified once from its reported ratios of deposits to total funding for 2014 to 2024. Deposit reliance is high if the median ratio is at least 0.25 and low otherwise. Lenders without usable observations are reported separately as unknown and remain in the estimation with separate shock interactions. Lenders is the number of lenders of each type in a group, and loans is the group's share of that type's loans in the estimation sample. A lender that changes type is counted in each type in which it appears.

4 Empirical Methodology

This section sets out the empirical model we use to estimate how the response of credit terms to a monetary policy shock differs across groups of lenders. Since each loan appears only once in our sample, indexed by origination month, the data form a pooled cross-section of loans rather than a panel. In an approach akin to a local projection (Jordà, 2005), we take advantage of the cross-sectional richness of the data to estimate the effect of a shock in month $t-h$ on the terms of loans originated in month t, for $h = 0, \ldots, 12$. We first present the estimating equation and the conditions under which its coefficients identify the effect of the shock, and then describe how we estimate it and conduct inference.

4.1 Estimating equation

To estimate the effects of monetary policy shocks on loan terms, we compare loans originated by different lenders at different points in time after a monetary policy shock. Unlike a standard local projection, this comparison does not follow the same borrower over time. Instead, to make loans comparable across periods, we use a rich set of fixed effects that control for the lender, the vehicle make, whether the vehicle is new or used, the borrower's employment status and type, and the loan product. In addition, because macroeconomic conditions change over time and may differ across countries, we also include country × month fixed effects. For each horizon $h = 0, \ldots, 12$, the underlying econometric model is

$$\begin{aligned} y_{i,j,c,t} = & \sum_{g \in \mathcal{G}} \beta_g^h \, D_{g,i} \, \widetilde{\Delta\rho}_{t-h} \\ & + \alpha_{c,t}^h + \alpha_j^h + \alpha_m^h + \alpha_n^h + \alpha_e^h + \alpha_u^h + \alpha_p^h + \varepsilon_{i,j,c,t}^h. \end{aligned} \tag{8}$$

In this specification, $y_{i,j,c,t}$ is the loan rate, in percent, or the log of the loan-to-value ratio, the maturity, or the loan amount of loan i, originated by lender j in country c in month t, and $\widetilde{\Delta\rho}_{t-h}$ is the monetary policy shock in month $t-h$ normalized as defined in Section 3.3. The indicator $D_{g,i}$ equals one if loan i belongs to group g, and the set of groups $\mathcal{G}$ excludes one reference group, so that β_g^h measures the effect of the shock on the terms of loans in group g, h months later, relative to loans in the reference group. The fixed effects are for the country and month of origination ($\alpha_{c,t}^h$), the lender (α_j^h), the vehicle make (α_m^h), new or used vehicles (α_n^h), the borrower's employment status in each country (α_e^h), the type of customer (α_u^h), and the loan product (α_p^h), and the superscript h indicates that every coefficient and fixed effect is specific to the horizon.

To make the estimated effects smoother across horizons, we implement the method of Barnichon and Brownlees (2019), which approximates the path of each β_g^h across horizons

by a combination of cubic B-spline functions of the horizon, $\beta_g^h = \sum_{k=1}^{K} \theta_{g,k}\, b_k(h)$, and penalizes the second differences of the coefficients $\theta_{g,k}$. Substituting this approximation into equation (8) gives the estimated equation

$$\begin{aligned} y_{i,j,c,t} = & \sum_{g \in \mathcal{G}} \sum_{k=1}^{K} \theta_{g,k}\, b_k(h)\, D_{g,i}\, \widetilde{\Delta \rho}_{t-h} \\ & + \alpha_{c,t}^h + \alpha_j^h + \alpha_m^h + \alpha_n^h + \alpha_e^h + \alpha_u^h + \alpha_p^h + \varepsilon_{i,j,c,t}^h, \end{aligned} \tag{9}$$

which we estimate jointly for all horizons $h = 0, \ldots, 12$ on a data set in which each loan appears once for each horizon.

The country $\times$ month fixed effects absorb the part of the shock's effect that is common to all loans originated in a given country and month. The model therefore identifies only differences in the effect of the shock across groups of lenders, which is what we need to compare how monetary policy transmits through different lenders. Moreover, because the monetary policy shock is uncorrelated both with its own leads and lags and with other shocks, each β_g^h captures the differential effect of the shock h months after it occurs.

4.2 Estimation and inference

We estimate the coefficients $\theta_{g,k}$ of equation (9) by penalized least squares on the stacked data,

$$\widehat{\theta}_\lambda = \arg\min_{\theta} \left\| M_F Y - M_F X \theta \right\|^2 + \lambda\, n\, \theta' P \theta, \tag{10}$$

where Y stacks the credit terms of all loans at all horizons, X stacks the regressors $b_k(h)$ $D_{g,i}\, \widetilde{\Delta \rho}_{t-h}$, M_F removes the horizon-specific fixed effects from both, n is the number of stacked observations, and P applies the second-difference penalty to the coefficients of each response path separately. The parameter λ sets how smooth the response paths are. We use $K = 15$ cubic B-splines and set λ to target 7.5 effective degrees of freedom per response path, compared with 13 for estimates of equation (8) horizon by horizon. Appendix B.3 reports estimates with targets of 5 and 10 effective degrees of freedom and with 10 and 20 B-splines.

Since all loans exposed to the same shock month share the same value of the shock, we compute standard errors that allow for arbitrary correlation of the errors across all loans within a shock month, as well as for serial correlation across shock months up to 12 lags, following Driscoll and Kraay (1998) with a Newey–West kernel (Newey and West, 1987). Following Barnichon and Brownlees (2019), we compute the confidence intervals with a smaller penalty, $\lambda/10$, than the one we use for the point estimates.

We estimate equation (9) on 1,831,887 loans originated by 15 lenders in Germany, Spain, France, Italy and Portugal between January 2014 and March 2025, keeping the loans for which all four credit terms, all the fixed effects and the shock at every horizon are observed, so that all outcomes are estimated on the same loans (Table 1). The three main specifications in Section 5 differ only in how they define the groups g and the reference group. Section 6 estimates the same equation on alternative samples and with alternative funding measures, shocks and smoothing settings.

5 Results

We apply the framework presented in Section 2 to the data described in Section 3. Following the order of the analysis suggested in Section 2.3: we first compare lenders by type, then by deposit reliance, and finally by both together. Each step answers a different question. The type comparison asks whether captives, specialized lenders and banks transmit the same shock differently. The funding comparison asks whether lenders that face a larger funding-cost shock, in the sense of equations (4) and (5), pass more of it on. The joint comparison asks whether the type differences survive once lenders with similar funding are compared with each other, which is the test the framework proposes for separating ownership from funding.

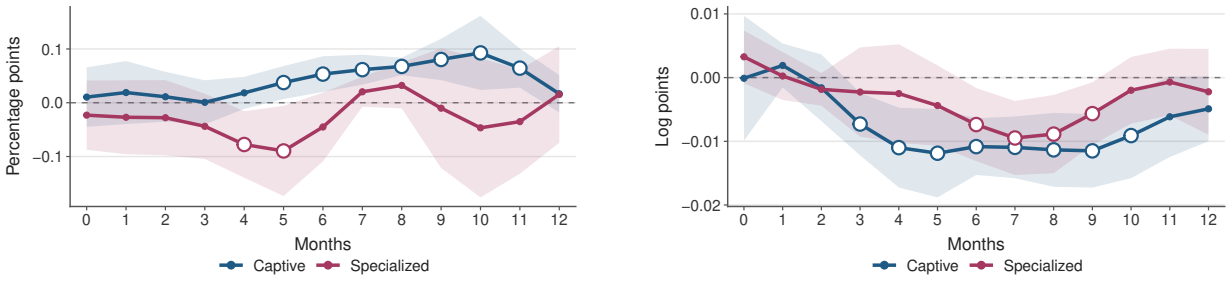
Figures 1–4 report the estimates of equation (9) for the four credit terms over the twelve months that follow a normalized monetary-policy surprise, constructed as in Section 3.3. Two features of the design, discussed in Section 4, shape how the figures should be read. First, country × origination-month fixed effects absorb the common response of the market, so every estimate is a difference in responses between groups of lenders lending in the same country and month, not the level of pass-through. A negative differential for a group therefore need not mean that its loan rates fell. Second, each observation is a separate loan indexed by its origination month, so a path compares successive origination cohorts rather than changes within an existing contract. Loan-rate differences are in percentage points and the other outcomes in changes in logs, and all intervals are pointwise 95 percent confidence intervals.

The levels of credit terms differ across lender types. Table 1 shows that captive loans carry lower rates than bank loans on average (5.51 versus 6.60 percent), but also have lower loan-to-value ratios and shorter maturities. These differences describe the terms and composition of observed loans. They do not determine how the same lenders respond when a common monetary-policy surprise changes funding costs. We therefore examine responses across all four margins.

5.1 Lender type

The framework allows lender type to matter for two separate reasons. Specialized lenders may originate and screen auto loans at lower cost ($a_S < a_B$), and captives value an additional borrower at $W_C = \mu_C + MD_C$ because they internalize the parent's profit on the vehicle. Neither feature implies a particular sign for the difference in responses to a funding shock. A view common in the literature, however, suggests that captives should insulate their borrowers. The captive is the only lender whose parent loses a sale when higher rates push a buyer out of the market, and captives have been shown to use credit to support sales and resale values (Murfin and Pratt, 2019; Bodnaruk et al., 2016). If this motive shaped their response to monetary policy, captive rates should rise by less than bank rates after a tightening.

Figure 1 does not support this view. Captive loan rates respond more positively than bank loan rates at months five to eleven (Figure 1a). The month-nine estimate is 0.080 percentage points, with a 95 percent interval of $[0.042, 0.119]$. The impact estimate is 0.010 $[-0.045, 0.066]$, and the interval includes zero again by month twelve. Captives also have lower loan-to-value responses than banks at months three to ten; the month-nine difference is -0.0115 $[-0.0173, -0.0057]$ log points (Figure 1b). Their maturity response is lower at months seven to ten and their loan-amount response at months four and six to ten. At month nine these differences are -0.0069 $[-0.0116, -0.0022]$ and -0.0143 $[-0.0247, -0.0039]$ log points. The four margins therefore point to tighter captive terms relative to banks at intermediate horizons. The impact differences in maturity and loan amount are imprecise, so the evidence does not establish an immediate adjustment on those margins.

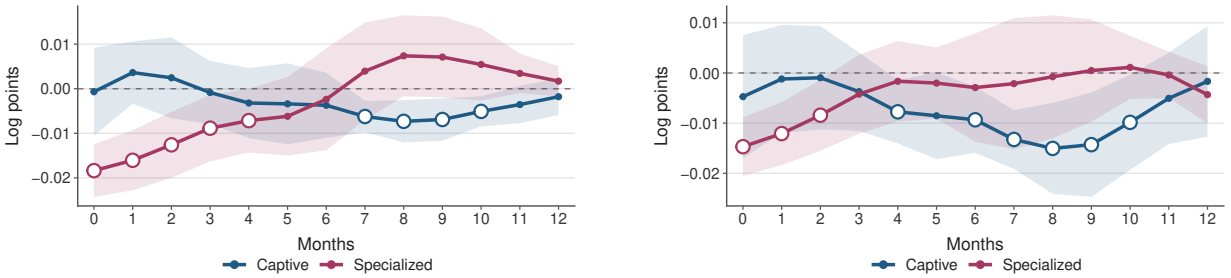


(a) Loan rate (b) Loan-to-value ratio

Figure 1: Responses of credit terms at captive lenders and specialized lenders relative to banks. Each panel plots the difference between the response of a credit term at captive lenders, or at specialized lenders, and its response at banks in the same country and origination month, h months after a normalized monetary-policy surprise, estimated from equation (9). Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero. Panels (c) and (d) are shown later in the text.

The framework separates these response differences from differences in levels. The manufacturing margin changes the captive's response to a funding shock through its market position and through the share D_C of its marginal borrowers who would otherwise not buy; neither channel fixes the sign of the response. Lower average captive loan rates therefore need not imply a smaller response to tightening. The nature of the shock also matters. Evidence from the collapse of asset-backed funding in 2008 (Benmelech et al., 2017), the Volkswagen emissions scandal (Benetton et al., 2025b), and metal tariffs (Hankins et al., 2026) concerns shocks directly linked to manufacturers or their funding. A common monetary policy surprise provides a different comparison. Our estimates establish differential credit terms among observed originations, without identifying the parent's incentive separately from other captive characteristics.

Specialized lenders have a different pattern across terms and horizons. Their loan-rate response is below the bank benchmark at months four and five, but the intervals include zero elsewhere. At month six, for example, the difference is -0.045 $[-0.109, 0.019]$ percentage points. Their loan-to-value response is also lower than banks' at months six to nine, with a month-six difference of -0.0074 $[-0.0131, -0.0016]$ log points. Maturities respond less at months zero to four and loan amounts at months zero to two: the impact differences are -0.0184 $[-0.0243, -0.0125]$ and -0.0147 $[-0.0205, -0.0089]$ log points. Thus the occasional lower rate differential does not establish that specialized lenders insulate borrowers across the contract. This matters in auto credit, where maturity as well as the interest rate affects the monthly payment (Attanasio et al., 2008; Argyle et al., 2020).

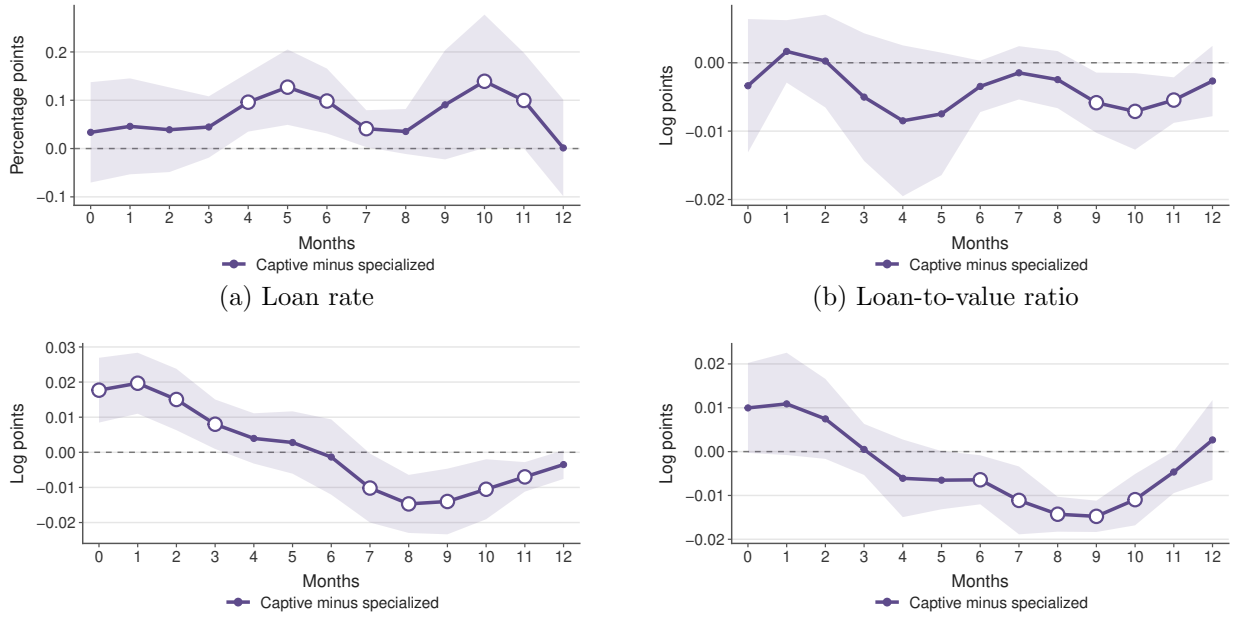


(c) Maturity (d) Loan amount

Figure 1: Responses of credit terms at captive lenders and specialized lenders relative to banks (continued). Panels (c) and (d) are changes in logs. Notes as in panels (a) and (b).

The direct captive–specialized contrasts in Figure 2 show how the two auto-focused groups differ. Captives have more positive rate responses at months four to seven and ten to eleven; at month six the gap is 0.098 $[0.031, 0.166]$ percentage points. Their loan-to-value responses are lower at months nine to eleven, with a month-nine difference of -0.0058 $[-0.0102, -0.0014]$ log points. Maturity differences change sign: captives have higher re-

sponses at months zero to three but lower responses at months seven to eleven. The month-nine maturity gap is -0.0140 $[-0.0234, -0.0047]$ log points. Loan-amount responses are lower at months six to ten. The relative adjustment therefore depends on the term and the horizon. These direct comparisons use the covariance between the two estimated paths; they do not identify the manufacturer's contribution to captive decisions in isolation.



(a) Loan rate (b) Loan-to-value ratio

(c) Maturity (d) Loan amount

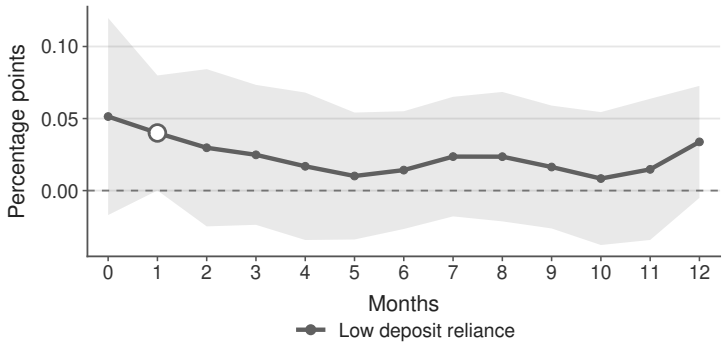
Figure 2: Difference between the responses of credit terms at captive lenders and at specialized lenders. Each panel plots the captive–bank differential minus the specialized–bank differential, both as shown in Figure 1, h months after a normalized monetary-policy surprise. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals for the difference, based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero.

5.2 Deposit funding

The framework makes a sharper prediction for funding than for lender type. Because deposit rates reprice more slowly than non-deposit funding ($\phi_D < \phi_W$ in equation (4)), a lender with a larger deposit share suffers a smaller direct cost effect after a tightening (equation (5)). If own-cost pass-through is positive, its loan rates should therefore rise by less. Because deposits catch up with the policy rate over time, the framework also predicts that this difference should be largest shortly after the shock. Figure 3 tests both predictions by grouping lenders on deposit reliance alone, regardless of type.

The rate estimates provide limited support for these predictions (Figure 3a). The low-minus-high-deposit difference is 0.051 $[-0.017, 0.120]$ percentage points on impact and 0.014

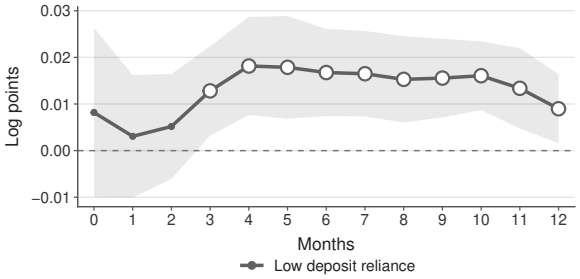
$[-0.027, 0.055]$ at month six. The point estimates are positive throughout, but the pointwise interval excludes zero only at month one, where its lower bound is close to zero. The data therefore remain compatible with a modest positive rate differential as well as with little difference at most horizons. They do not establish either a persistent funding gradient in rates or a clear sequence in which an initial funding advantage disappears as deposits reprice. The deposits channel of Drechsler et al. (2017, 2021) motivates this comparison, but the estimated rate path alone is weak evidence for it in this sample.



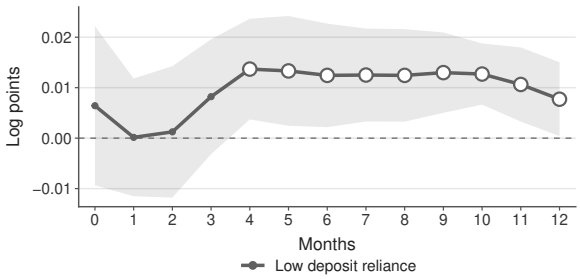
(a) Loan rate

Figure 3: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance. Each panel plots the difference between the response of a credit term at lenders with low deposit reliance and its response at lenders with high deposit reliance in the same country and origination month, h months after a normalized monetary-policy surprise, estimated from equation (9). Lenders are grouped by deposit reliance alone, regardless of type. Deposit reliance, defined in Section 3, is high if the lender's median ratio of deposits to total funding over 2014–2024 is at least 0.25 and low if it is below 0.25. Lenders with unknown deposit reliance are retained with separate shock interactions. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero. Panels (b) to (d) are shown later in the text.

The funding differences are clearer in non-rate terms. Low-deposit lenders have higher loan-to-value responses at months three to twelve and higher maturity responses at months four to twelve. At month six, the loan-to-value difference is 0.0167 $[0.0074, 0.0261]$ log points; the maturity difference is 0.0124 $[0.0022, 0.0227]$ (Figures 3b and 3c). These are relative responses, not evidence that loan-to-value ratios or maturities rise in absolute terms. Studies of euro-area pass-through such as Altavilla et al. (2020), Vilerts et al. (2025), and De Jonghe et al. (2026) emphasize lending rates; our results show that funding-related heterogeneity can be clearer in other terms. The signs are consistent with lenders adjusting contract design to remain competitive, particularly when borrowers respond to monthly payments (Attanasio et al., 2008; Argyle et al., 2020). They do not directly measure that mechanism. As the framework allows, the joint response of contract terms can differ from the fixed-attractiveness funding-exposure comparison in Appendix A.4.



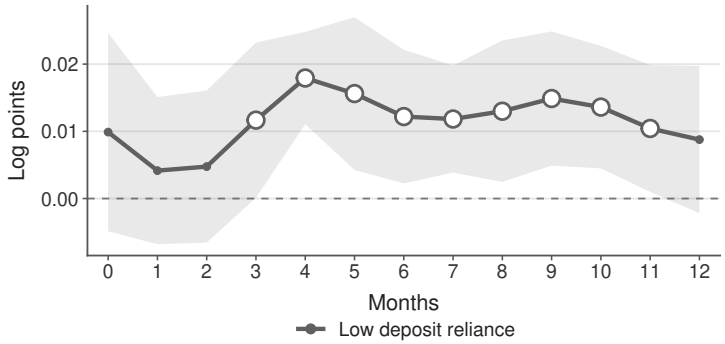
(b) Loan-to-value ratio



(c) Maturity

Figure 3: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance (continued). Panels (b) and (c) are changes in logs. Notes as in panel (a).

Loan-amount responses are also higher at low-deposit lenders at months three to eleven (Figure 3d). At month six the difference is 0.0122 $[0.0023, 0.0221]$ log points; the intervals include zero on impact and at month twelve. The estimates concern amounts per observed loan and do not establish how many borrowers receive credit. This distinction limits comparisons with evidence on aggregate substitution between banks and nonbanks after tightening (Xiao, 2020; Elliott et al., 2024; Buchak et al., 2024; Cucic and Gorea, 2026). Because each funding group contains several lender types, the funding-only specification also leaves ownership and specialization mixed into the comparison. We turn to the joint specification to compare lender types within funding categories.



(d) Loan amount

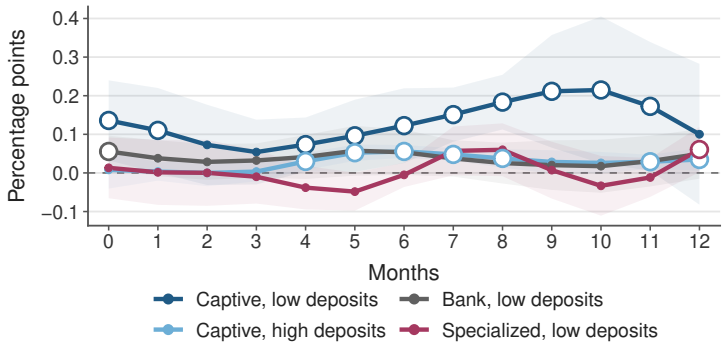
Figure 3: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance (continued). Panel (d) is in changes in logs. Notes as in panel (a).

5.3 Lender type and deposit funding

Section 2.3 proposed a simple test for separating funding from ownership. If the differences between lender types disappear once lenders with similar funding are compared, funding exposure is the main source of heterogeneity. If they remain, funding alone is not enough. This

test is not available in the literature that compares banks with nonbank lenders (Xiao, 2020; Buchak et al., 2024; Elliott et al., 2024; Cucic and Gorea, 2026; Haque et al., 2025; Elliott et al., 2025), because there the nonbank is, almost by definition, the lender without deposits, so funding and the other features of nonbanks move together. Our setting differs because funding varies within lender type: several captives hold banking licences and take deposits, and several banks rely mainly on non-deposit funding (Table 3). The joint specification exploits this variation and uses high-deposit banks as the reference group (Figure 4).

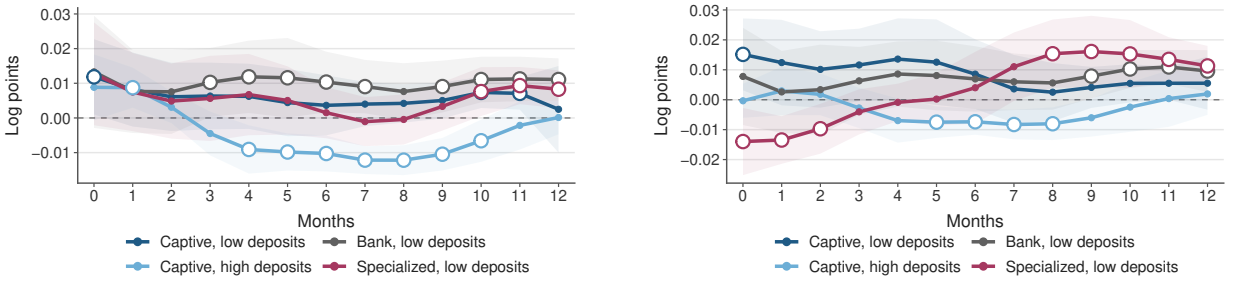
The loan-rate panel shows that the captive differential remains within the low-deposit group (Figure 4a). For low-deposit captives, the impact difference relative to high-deposit banks is 0.136 $[0.032, 0.240]$ percentage points. At month nine it is 0.211 $[0.066, 0.357]$. Low-deposit banks also have a positive impact differential, at 0.056 $[0.015, 0.096]$, but their later intervals include zero. The direct low-deposit captive–bank contrast is positive at months six to ten; at month nine it is 0.191 $[0.063, 0.320]$ percentage points. Thus differences in the deposit category do not account for the later captive–bank rate differential. High-deposit captives also have more positive rate responses than high-deposit banks at months four to eight and eleven to twelve. These comparisons are consistent with lender characteristics beyond deposit reliance shaping transmission, but they do not identify ownership incentives separately from other differences between the institutions.



(a) Loan rate

Figure 4: Responses of credit terms by lender type and deposit reliance relative to banks with high deposit reliance. Each panel plots the difference between the response of a credit term at a group of lenders and its response at banks with high deposit reliance in the same country and origination month, h months after a normalized monetary-policy surprise, estimated from equation (9). The plotted groups are captive lenders with low deposit reliance, captive lenders with high deposit reliance, banks with low deposit reliance, and specialized lenders with low deposit reliance; no specialized lender has high deposit reliance. Deposit reliance, defined in Section 3, is high if the lender's median ratio of deposits to total funding over 2014–2024 is at least 0.25 and low if it is below 0.25. Lenders with unknown deposit reliance are retained with separate shock interactions. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero. Panels (b) to (d) are shown later in the text.

Funding differences also appear within captives. The direct low-minus-high-deposit rate contrast is positive at months seven to nine, reaching 0.183 $[0.044, 0.322]$ percentage points at month nine. Its month-six interval includes zero, at 0.067 $[-0.022, 0.156]$. For non-rate terms, low-deposit captives have higher loan-to-value responses than high-deposit captives at months three to eleven; their maturity response is higher on impact and at months three to six. At month six these differences are 0.0139 $[0.0040, 0.0238]$ and 0.0159 $[0.0040, 0.0278]$ log points. Their loan-amount response is higher at months zero to six. These are direct within-captive contrasts, not conclusions drawn from different significance levels against the bank reference group. They show that captive ownership does not imply a common response across funding models.



(b) Loan-to-value ratio

(c) Maturity

(d) Loan amount

Figure 4: Responses of credit terms by lender type and deposit reliance relative to banks with high deposit reliance (continued). Panels (b) to (d) are changes in logs. Notes as in panel (a).

The joint estimates also distinguish comparisons within a funding group from comparisons against high-deposit banks. Low-deposit captives have lower loan-to-value responses than low-deposit banks at months six, seven, and nine, and lower loan-amount responses at months seven to eleven. Their maturity differences change sign and are imprecise at most horizons. Together with the rate contrast, these results indicate that measured deposit reliance does not account for every difference between captives and banks. They do not support a simple division in which funding determines the speed of rate adjustment and ownership determines the later response. The reference group contains three banks, and there are no high-deposit specialized lenders, so some comparisons rest on few institutions. Differences in borrower

composition, other lender characteristics, and selection into securitization remain possible explanations for part of the observed heterogeneity. Section 6 examines how far these results depend on the composition of the securitized sample, the definition of funding and the construction of the shock.

6 Robustness

This section re-estimates equation (9) with changes to the sample, the funding measure, the shock and the smoothing. Each check keeps the fixed effects and inference of Section 4, and Appendix B reports its figures and tables. The captive–bank rate differential stays positive at intermediate horizons under each alternative sample, funding measure, shock and smoothing setting. It is shorter-lived with the total surprise and imprecise among loans that enter pools late. The funding differentials in loan-to-value ratios, maturities and loan amounts also persist, except with the total surprise. None of the checks measures the response of aggregate auto credit or shows that securitized loans represent all auto lending.

6.1 Pool entry and sample composition

Loans enter securitization pools with a lag that varies across loans and lenders. In the estimation sample the shortest lag is 14 days and the median is 584 days (Table 4). Klee and Shin (2020) find that US auto lenders hold higher-quality loans longer before securitizing them, so the lag can also reflect which loans lenders choose to securitize. Figure 5 uses the log number of days from origination to pool entry as the outcome. Relative to banks, specialized lenders' loans enter pools sooner at months zero to five after the shock; at month four the difference is -0.097 $[-0.158, -0.037]$ log points. The captive intervals include zero at every horizon. Securitization timing therefore responds to the shock for some lenders.

We split loans at the median lag into fast and slow entry, and into quartiles at 311, 584 and 1,029 days (Table 4). Each lender-type differential uses banks in the same entry group as its benchmark. Among fast-entry loans, the captive rate differential is positive at months four to eleven and reaches 0.097 $[0.059, 0.135]$ percentage points at month nine, against 0.080 in the full sample (Figure 6). Among slow-entry loans the point estimates are also positive, at 0.071 at month nine, but the intervals include zero at every horizon. Captive loan-to-value, maturity and loan-amount differentials are negative at intermediate horizons in both groups. By quartile, the captive rate differential is positive at months six to eleven for the fastest-entry loans, at fewer horizons in the second and third quartiles, and at none in the fourth (Figure 7). The specialized rate differentials change sign across quartiles (Figure 8).

The grouping uses an outcome observed after origination and does not isolate exogenous variation in the lag. The bands test each bank-relative path separately and not equality across groups.

Lender continuity. Lenders that enter and leave the sample could mix changes in composition with responses to the shock. Every country–lender pair in the estimation sample originates loans in at least 91 percent of the calendar months between its first and last origination month. Restricting the sample to pairs above 75 or 90 percent of those months therefore removes no loans and leaves every estimate unchanged.

Excluding Germany. Germany has 12 percent of the loans but three of the six captive lenders (Table 2). Excluding it leaves 1,618,619 loans in Spain, France, Italy and Portugal (Figures 9–11). The captive–bank rate differential remains positive at months five to eleven, at 0.060 $[0.025, 0.094]$ percentage points at month six, compared with 0.053 in the full sample. Low-deposit lenders keep higher loan-to-value, maturity and loan-amount responses at intermediate horizons, with a month-six loan-to-value difference of 0.0214 $[0.0097, 0.0332]$ log points, and their rate intervals include zero at every horizon. In the joint specification, the direct low-deposit captive–bank rate contrast is positive at months six to ten (0.198 $[0.042, 0.353]$ percentage points at month nine). Low-deposit captives also keep higher loan-to-value responses than high-deposit captives at months three to eleven.

6.2 Funding and shock definitions

As an alternative to deposit reliance, we classify a lender as deposit taking if it reports positive deposits in any year from 2014 to 2024 (Table 5). This grouping puts 93 percent of captive loans in the deposit-taking group, leaves two captives that take no deposits with 2.3 percent of captive loans, and adds one specialized lender, with 76 percent of specialized loans, to the deposit takers. The funding-only estimates barely change (Figure 12). Lenders without deposits again have higher responses in all three non-rate terms at intermediate horizons; at month six the loan-to-value and maturity differences are 0.0166 $[0.0074, 0.0259]$ and 0.0124 $[0.0022, 0.0226]$ log points. In the joint specification (Figure 13), deposit-taking captives have higher rate responses than deposit-taking banks at months five to eleven, at 0.125 $[0.044, 0.206]$ percentage points at month nine. The captive–bank rate differential therefore also appears among lenders with the same deposit status. Captives that take no deposits have higher rate responses than banks that take none at months nine and ten. Within captives, however, those without deposits have lower loan-to-value responses than deposit takers at months seven to twelve, the opposite sign to the baseline within-captive

contrast. This comparison rests on two lenders with 20,154 loans, and we do not draw conclusions about funding within captives from it. Neither funding measure captures reserve holdings or deposit-rate rigidity. Among German banks selling securities during quantitative easing, high-deposit banks facing deposit-rate floors retain more reserves and lend relatively less to firms (Bittner et al., 2026). Reserve-rich banks' credit supply responds less to the ECB's 2022 tightening (Fricke et al., 2026).

The alternative shocks repeat all three main specifications. The total surprise keeps the information component that Section 3.3 removes. The equal-weight surprise averages the pure surprises over the calendar days of the month, so each announcement receives the same weight whatever its date. Each series is divided by its standard deviation in the reference sample that gives the baseline divisor of 0.0504733: 0.0785759 for the total surprise and 0.0358613 for the equal-weight surprise. These comparisons therefore assess shock construction and not the response to an identical change in rates. With the total surprise, the captive–bank rate differential is positive at months eight and nine, at 0.086 $[0.028, 0.144]$ percentage points at month nine, and the captive loan-to-value differential is negative at months nine and ten (Figures 14–16). The funding-only responses become imprecise: the loan-to-value and maturity intervals exclude zero only at months ten and eleven, and the rate and loan-amount intervals include zero throughout. In the joint specification, the direct low-deposit captive–bank rate contrast is positive at months nine to eleven (0.230 $[0.063, 0.397]$ percentage points at month nine), while the within-captive loan-to-value contrast is imprecise throughout. The total surprise also carries news about the economic outlook, so it changes the economic content of the shock. Equal weighting preserves the baseline patterns (Figures 17–19). The captive rate differential is positive at months five to twelve, the funding differentials in loan-to-value ratios, maturities and loan amounts are positive from month three or four onward, and the direct low-deposit captive–bank rate contrast is positive on impact and at months six to eleven.

6.3 Estimator choices

Figures 20 and 21 re-estimate the lender-type specification with targets of 5 and 10 effective degrees of freedom per response path, and with bases of 10 and 20 cubic B-splines at the baseline target of 7.5. Under every setting the captive rate interval excludes zero at months five to eleven, and the month-nine estimate lies between 0.074 and 0.082 percentage points, against 0.080 in the baseline. The captive paths for the other terms and the specialized paths also stay close to the baseline. The specialized rate intervals include zero at all but a few horizons under every setting, and at all horizons with 5 effective degrees of

freedom. The smoothing choices therefore do not drive the estimates, although the sample and measurement qualifications above still apply.

7 Conclusion

We ask whether the type of lender and the source of funding matter for how monetary policy shocks are transmitted to credit terms. To answer this question we use data from the euro area auto loan securitization market, together with a measure of monetary policy surprises and lender balance sheet data. The euro area auto loan market is a good place to answer this question, as it is served by general banks, manufacturer-owned captive lenders, and specialized auto lenders, and lenders of the same type can differ in how they are funded, with some relying more on deposits and others more on non-deposit funding, such as wholesale markets or their parent group. With these data and setting we can compare lenders of the same type that are funded differently and lenders of different types that are funded in a similar way.

We find that both the type of lender and its funding matter, and that they matter in different ways. Captive lenders become more restrictive than other lenders after a tightening. They raise loan rates by more than general banks, also when compared with banks with similar funding, and they lower loan-to-value ratios, shorten maturities, and lend smaller amounts. Specialized lenders tighten the same non-price terms, but their loan rates move with those of banks. Lenders that rely more on non-deposit funding go the other way on non-price terms. They set loan rates close to those of deposit-financed lenders but finance more of the car, for longer, and lend larger amounts, and the same pattern in non-price terms holds when comparing captive lenders with low and high deposit reliance.

We also show that heterogeneity in transmission can run deeper than the labels usually attached to lenders. Lenders of the same type can respond differently depending on how they are funded, and lenders with similar funding can respond differently depending on who owns them and what they specialize in. Lenders whose funding costs go up by more than those of their competitors seem to compete on the terms that matter most for the monthly payment, which our framework allows for through competition between lenders. While captives have more reason than any other lender to keep car buyers in the market, we find little evidence that they protect their borrowers from monetary policy. On the contrary, we find that captive lenders tighten both price and non-price credit terms by more than other lenders.

References

Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Altavilla, C., Canova, F., and Ciccarelli, M. (2020). Mending the broken link: Heterogeneous bank lending rates and monetary policy pass-through. *Journal of Monetary Economics*, 110:81–98.

Andrade, P. and Ferroni, F. (2021). Delphic and odyssean monetary policy shocks: Evidence from the euro area. *Journal of Monetary Economics*, 117:816–832.

Argyle, B. S., Nadauld, T. D., and Palmer, C. J. (2020). Monthly payment targeting and the demand for maturity. *The Review of Financial Studies*, 33(11):5416–5462.

Attanasio, O. P., Goldberg, P. K., and Kyriazidou, E. (2008). Credit constraints in the market for consumer durables: Evidence from micro data on car loans. *International Economic Review*, 49(2):401–436.

Barnichon, R. and Brownlees, C. (2019). Impulse response estimation by smooth local projections. *The Review of Economics and Statistics*, 101(3):522–530.

Barron, J. M., Chong, B.-U., and Staten, M. E. (2008). Emergence of captive finance companies and risk segmentation in loan markets: Theory and evidence. *Journal of Money, Credit and Banking*, 40(1):173–192.

Benetton, M., Gavazza, A., and Surico, P. (2025a). Mortgage pricing and monetary policy. *American Economic Review*, 115(3):823–863.

Benetton, M., Mayordomo, S., and Paravisini, D. (2025b). Financial shocks and credit dislocation: The role of integrated manufacturer-lenders. Ssrn working paper, Social Science Research Network. Version dated 8 April 2025. Uses European Data Warehouse auto ABS loan-level data; Volkswagen emissions-scandal design. Earlier versions circulated as "Credit Fire Sales: Captive Lending as Liquidity in Distress" (2020) and "Dislocated Credit: Captive Lending as Liquidity Management".

Benmelech, E., Meisenzahl, R. R., and Ramcharan, R. (2017). The real effects of liquidity during the financial crisis: Evidence from automobiles. *The Quarterly Journal of Economics*, 132(1):317–365.

Bernanke, B. S. and Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4):27–48.

Beyene, W., Falagiarda, M., Ongena, S., and Scopelliti, A. (2026). Do lenders price diesel risk? Evidence from Dieselgate and low-emission zones in captive vs. independent banks. Working Paper Series 3228, European Central Bank.

Bittner, C., Rodnyansky, A., Saidi, F., and Timmer, Y. (2026). Mixing quantitative easing and interest rate policies at the effective lower bound: micro evidence from the euro area. *Review of Finance*. Advance article; corrected proof dated 12 October 2026.

Bodnaruk, A., O'Brien, W., and Simonov, A. (2016). Captive finance and firm's competitiveness. *Journal of Corporate Finance*, 37:210–228.

Brennan, M. J., Maksimovic, V., and Zechner, J. (1988). Vendor financing. *The Journal of Finance*, 43(5):1127–1141.

Buchak, G., Matvos, G., Piskorski, T., and Seru, A. (2024). Beyond the balance sheet model of banking: Implications for bank regulation and monetary policy. *Journal of Political Economy*, 132(2):616–693.

Campbell, J. R., Evans, C. L., Fisher, J. D. M., and Justiniano, A. (2012). Macroeconomic effects of Federal Reserve forward guidance. *Brookings Papers on Economic Activity*, 2012(1):1–80.

Cucic, D. and Gorea, D. (2026). Nonbank lending and the transmission of monetary policy. *The Review of Financial Studies*, 39(4):966–1014. Earlier version: BIS Working Papers No. 1211, September 2024.

De Jonghe, O., Beņkovskis, K., Bielskis, K., Bonfim, D., Bottero, M., Briglevics, T., Cesnak, M., Dirma, M., Emiris, M., Filep-Mosberger, P., Jouvanceau, V., Kaiser, N., Khametshin, D., Lalinský, T., Grolmusz, V. M., Moretti, L., Nikitins, A. J., Nunnari, A., Rodriguez-Moreno, M., Stefanova, E., Szabó, L. T., Vilerts, K., and Zhao, S. E. (2026). Household borrowing and monetary policy transmission: Post-pandemic insights from nine European credit registers. *International Journal of Central Banking*, 22(2):91–194.

Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.

Drechsler, I., Savov, A., and Schnabl, P. (2021). Banking on deposits: Maturity transformation without interest rate risk. *Journal of Finance*, 76(3):1091–1143.

Driscoll, J. C. and Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of economics and statistics*, 80(4):549–560.

Elliott, D., Meisenzahl, R. R., and Peydró, J.-L. (2024). Nonbank lenders as global shock absorbers: Evidence from US monetary policy spillovers. *Journal of International Economics*, 149:103908.

Elliott, D., Meisenzahl, R. R., Peydró, J.-L., and Turner, B. C. (2025). Nonbanks, banks, and monetary policy: U.S. loan-level evidence since the 1990s. Working Paper 2022-27, Federal Reserve Bank of Chicago. Revised 12 September 2025. Accepted at the Journal of Monetary Economics under the title "Nonbanks, Banks, and Monetary Policy: Real Effects across Three Major U.S. Credit Markets from Granular Data".

European DataWarehouse (2025a). Auto loans vs auto leases: Captives vs. non-captives. Accessed 27 June 2026.

European DataWarehouse (2025b). Data availability report Q4 2024. Accessed 27 June 2026.

European Securities and Markets Authority (2023). Questions and answers on the securitisation regulation. Technical Report ESMA33-128-563, European Securities and Markets Authority. Updated 13 July 2023.

Fliegel, P., Hagen, A., Koch, N., and Ritter, N. (2025). Carbon pricing and household finance: How banks price transition risk in auto loans. CESifo Working Paper 12288, CESifo.

Fricke, D., Greppmair, S., and Paludkiewicz, K. (2026). Excess reserves and monetary policy tightening. *Review of Finance*, 30(5):1777–1816.

Giuzio, M., Kaufmann, C., Ryan, E., and Cappiello, L. (2021). Investment funds, risk-taking, and monetary policy in the euro area. Working Paper 2021/2605, European Central Bank.

Hankins, K. W., Momeni, M., and Sovich, D. (2026). Consumer credit and the incidence of tariffs: Evidence from the auto industry. *American Economic Review*, 116(2):627–673.

Haque, S., Jang, Y. S., and Wang, J. J. (2025). Indirect credit supply: How bank lending to private credit shapes monetary policy transmission. FEDS 2025-059, Board of Governors of the Federal Reserve System.

Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises—the role of information shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Jiang, E. X., Matvos, G., Piskorski, T., and Seru, A. (2020). Which banks are (over) levered? insights from shadow banks and uninsured leverage. Working Paper 26903, National Bureau of Economic Research. Earlier versions circulated under the title "Banking Without Deposits.".

Jiménez, G., Ongena, S., Peydró, J.-L., and Saurina, J. (2012). Credit supply and monetary policy: Identifying the bank balance-sheet channel with loan applications. *American Economic Review*, 102(5):2301–2326.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Kashyap, A. K. and Stein, J. C. (2000). What do a million observations on banks say about the transmission of monetary policy? *American Economic Review*, 90(3):407–428.

Kilian, L. (2024). How to construct monthly var proxies based on daily surprises in futures markets. *Journal of Economic Dynamics and Control*, 168:104966.

Klee, E. and Shin, C. (2020). Post-crisis signals in securitization: Evidence from auto ABS. Finance and Economics Discussion Series 2020-042, Board of Governors of the Federal Reserve System.

Mimun, N. T., Fukker, G., and Sydow, M. (2025). The effects of monetary policy on banks and non-banks in times of stress. Technical Report 3114, ECB.

Miranda-Agrippino, S. and Ricco, G. (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics*, 13(3):74–107.

Murfin, J. and Pratt, R. (2019). Who finances durable goods and why it matters: Captive finance and the coase conjecture. *The Journal of Finance*, 74(2):755–793.

Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *The Quarterly Journal of Economics*, 133(3):1283–1330.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In Taylor, J. B. and Uhlig, H., editors, *Handbook of Macroeconomics*, volume 2A, chapter 2, pages 71–162. Elsevier, Amsterdam.

Vilerts, K., Anyfantaki, S., Beņkovskis, K., Bredl, S., Giovannini, M., Horky, F. M., Kunzmann, V., Lalinský, T., Lampousis, A., Lukmanova, E., Petroulakis, F., and Zutis, K. (2025). Details matter: Loan pricing and transmission of monetary policy in the euro area. Working Paper Series 3078, European Central Bank.

Xiao, K. (2020). Monetary transmission through shadow banks. *The Review of Financial Studies*, 33(6):2379–2420.

Yao, V. and Wang, Q. (2024). Fintech, market power and monetary transmission. Circulating working paper (SSRN).

A Analytical Details of the Conceptual Framework

This appendix makes the demand system, lending profits, captive objective, funding exposure, and local transmission results in Section 2 explicit.

A.1 Demand

For analytical convenience, suppose a potential financed buyer chooses among the three lending channels and the option not to make the financed purchase. Utility from financing through lender j is

$$\xi_j + v(z_j) + \varepsilon_{ij},$$

where ξ_j captures preferences for a channel unrelated to contract terms, such as relationships, convenience, or dealer arrangements. Utility from the outside option is ε_{i0}. The idiosyncratic terms are independent type-I extreme value draws. The resulting logit shares are

$$s_j(z) = \frac{\exp(\xi_j + v(z_j))}{1 + \sum_{k\in\{B,S,C\}} \exp(\xi_k + v(z_k))}, \qquad s_0(z) = \frac{1}{1 + \sum_{k\in\{B,S,C\}} \exp(\xi_k + v(z_k))}. \quad (11)$$

The index v is decreasing in the loan rate and may depend jointly on the rate, LTV, and maturity. Other smooth demand systems would change the first-order conditions only through the corresponding demand derivatives.

A.2 Lending profit

The initial principal of a loan is ℓP. Let $b_z(t)$ denote its outstanding performing balance at time t, and let τ_z be the time of prepayment or default, with $\tau_z = \infty$ if neither occurs before maturity. At discount rate δ, define

$$H(z) = \mathbb{E}\left[\int_0^{\min\{T,\tau_z\}} e^{-\delta t} b_z(t)\, dt\right]. \quad (12)$$

Thus $H(z)$ is the expected discounted balance that must be funded over the life of the loan; it can change with the initial balance, maturity, amortization, prepayment, and default.

Let $R(z)$ denote expected discounted lending returns before funding and screening costs, including loan revenues net of expected credit losses and servicing costs. Per-loan lending profit is

$$\mu_j(z_j; f_j) = R(z_j) - f_j H(z_j) - a_j. \quad (13)$$

The specialized lender's screening and origination cost satisfies $a_S < a_B$, while a_C is unrestricted.

A.3 The captive

First consider any smooth differentiated-demand system in which lender j's contract affects demand through a single index $v_j = v(z_j)$. For the captive, define

$$D_C \equiv \frac{dQ/dv_C}{ds_C/dv_C}. \tag{14}$$

For any element x of the captive's contract,

$$\frac{\partial \Pi_C}{\partial x} = s_C \frac{\partial \mu_C}{\partial x} + \left(\frac{ds_C}{dv_C} \mu_C + M \frac{dQ}{dv_C} \right) \frac{\partial v_C}{\partial x} = s_C \frac{\partial \mu_C}{\partial x} + \frac{ds_C}{dv_C} \left(\mu_C + M D_C \right) \frac{\partial v_C}{\partial x}.$$

At any given profile of offers, this is the same first-order condition as for an independent lender whose per-loan cost is $a_C - M D_C$. Thus the manufacturing term acts like a reduction in the captive's per-loan cost by the value of the additional vehicle sales generated at the margin.

Under the logit demand system in equation (11),

$$D_C = \frac{s_0}{1 - s_C} = \frac{1}{1 + \exp(\xi_B + v(z_B)) + \exp(\xi_S + v(z_S))}. \tag{15}$$

This expression falls in $v(z_B)$ and $v(z_S)$, so it rises when rival offers become less attractive.

The logit first-order conditions for the full contract can be written as

$$0 = s_j \nabla \mu_j + \left[s_j (1 - s_j) \mu_j + \mathbf{1}\{j = C\} M s_j s_0 \right] \nabla v(z_j). \tag{16}$$

Equivalently,

$$\nabla \mu_j + (1 - s_j) W_j \nabla v(z_j) = 0,$$

where

$$W_B = \mu_B, \qquad W_S = \mu_S, \qquad W_C = \mu_C + M D_C. \tag{17}$$

A.4 Funding exposure

Holding borrower attractiveness fixed at $v(z) = u$, shares and vehicle sales are fixed, so the screening cost a_j and the manufacturing term do not vary across contracts. Every lender, including the captive, therefore prefers the contract that maximizes $R(z) - fH(z)$ among

those with $v(z) = u$. Let z and z' solve this problem at funding costs $f < f'$. Optimality implies

$$R(z) - fH(z) \geq R(z') - fH(z')$$

and

$$R(z') - f'H(z') \geq R(z) - f'H(z).$$

Adding the inequalities gives

$$(f' - f)\{H(z) - H(z')\} \geq 0,$$

so $H(z') \leq H(z)$. At a fixed level of borrower attractiveness, a higher funding cost therefore weakly reduces the funding exposure of the preferred contract. This does not require LTV and maturity to both fall.

A.5 Transmission

We assume a regular interior equilibrium: each lender's contract is a strict local best response, and the Jacobian of the stacked first-order conditions is nonsingular. By the implicit-function theorem, equilibrium contracts are then differentiable in funding costs, and equation (6) follows from the chain rule.

The direct cost effect is the change in lending profit at a fixed contract. Since $df_j/d\rho = \phi_j$,

$$\frac{\partial \mu_j}{\partial \rho} = -H(z_j)\phi_j. \tag{18}$$

A larger ϕ_j therefore produces a larger decline in per-loan profit before the lender adjusts its contract.

A controlled comparison of funding structures varies lender k's deposit share while holding its reference funding cost and all other primitives fixed. At $\rho = \bar{\rho}$, this changes ϕ_k without changing the reference equilibrium. Therefore,

$$\frac{\partial}{\partial d_k}\left(\frac{dz_j^*}{d\rho}\right) = (\phi_D - \phi_W)\frac{\partial z_j^*}{\partial f_k}. \tag{19}$$

Because $\phi_D < \phi_W$, a higher deposit share attenuates the contribution of lender k's funding-cost shock. If lender k's own-cost pass-through, $\partial r_k^*/\partial f_k$, is positive, a higher deposit share reduces the contribution of its own funding-cost shock to the equilibrium loan-rate response. No analogous sign is imposed on each non-price term or on comparisons across lenders whose equilibrium responses differ.

B Robustness exhibits

This appendix reports the tables and figures for Section 6. Each figure re-estimates equation (9) with the fixed effects, smoothing and inference of the baseline unless its caption states otherwise.

Table 4: Time from origination to pool entry

			Days from origination to pool entry		
	Loans	Share (%)	Minimum	Median	Maximum
Fast entry	916,169	50.0	14	311	584
Slow entry	915,718	50.0	585	1,029	3,532
Quartile 1 (fastest)	459,332	25.1	14	205	311
Quartile 2	456,837	24.9	312	419	584
Quartile 3	458,454	25.0	585	792	1,029
Quartile 4 (slowest)	457,264	25.0	1,030	1,341	3,532

Notes: Fast-entry loans enter a securitization pool within 584 days of origination, the median, and slow-entry loans later. The quartiles split at 311, 584 and 1,029 days. The sample is that of Table 1.

Table 5: Lenders by deposit taking

	Takes deposits		No deposits		Unknown	
	Lenders	Loans (%)	Lenders	Loans (%)	Lenders	Loans (%)
Banks	4	61.2	3	38.8	0	0.0
Captive lenders	3	92.8	2	2.3	1	4.9
Specialized lenders	1	75.8	3	24.2	0	0.0

Notes: A lender takes deposits if it reports positive deposits in any year from 2014 to 2024. Lenders without usable observations are reported separately as unknown and remain in the estimation with separate shock interactions. Lenders is the number of lenders of each type in a group, and loans is the group's share of that type's loans in the estimation sample. A lender that changes type is counted in each type in which it appears. Table 3 gives the baseline grouping by deposit reliance.

B.1 Pool entry and sample composition

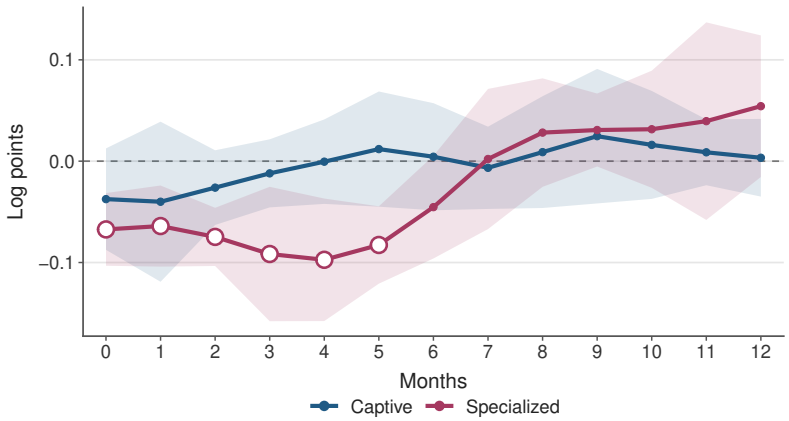
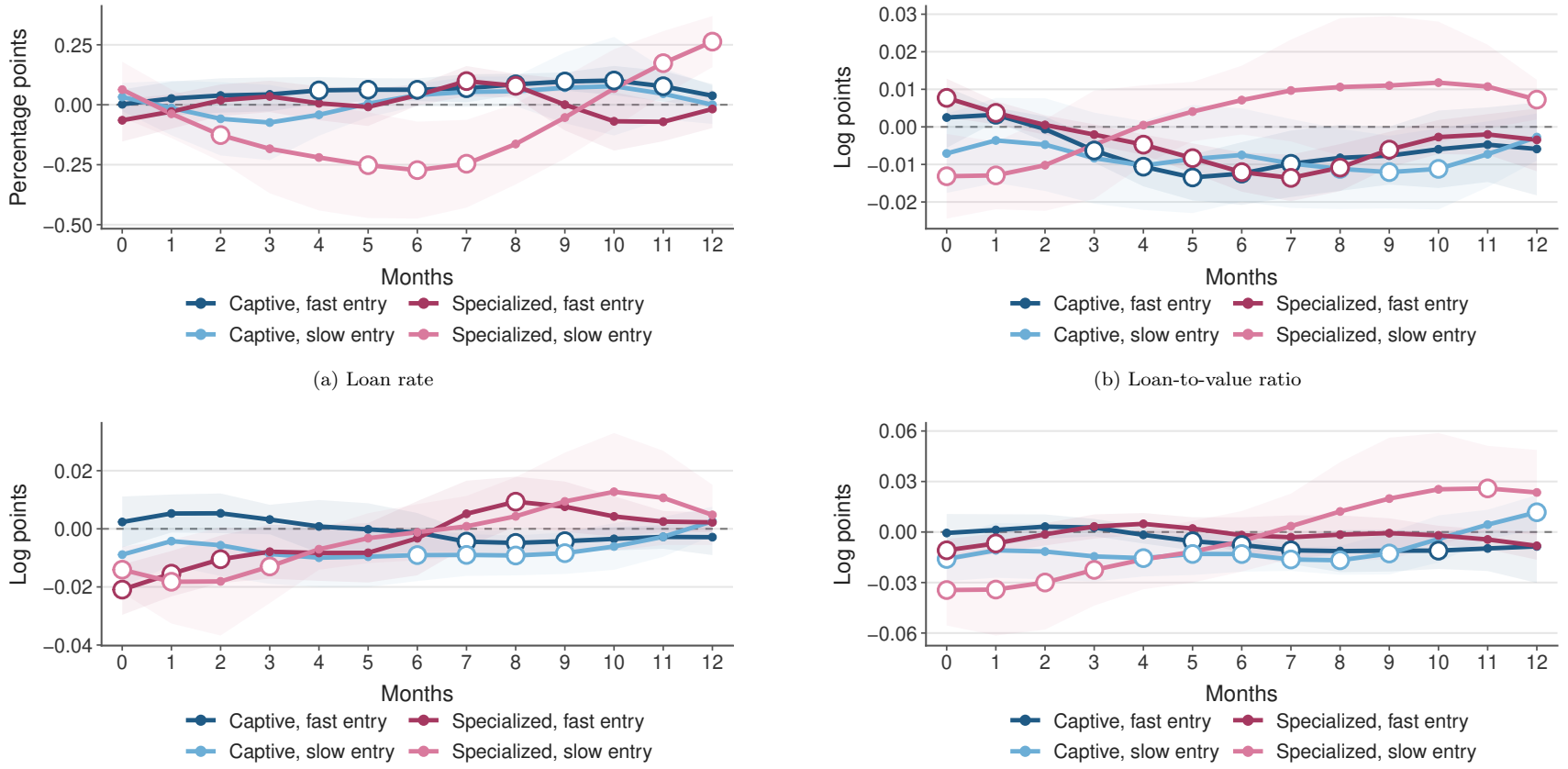


Figure 5: Response of the time to pool entry at captive lenders and specialized lenders relative to banks. The figure plots the difference between the response of the log number of days from origination to entry into a securitization pool at captive lenders, or at specialized lenders, and its response at banks in the same country and origination month, h months after a normalized monetary-policy surprise, estimated from equation (9) with this outcome. The vertical axis is in changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero.



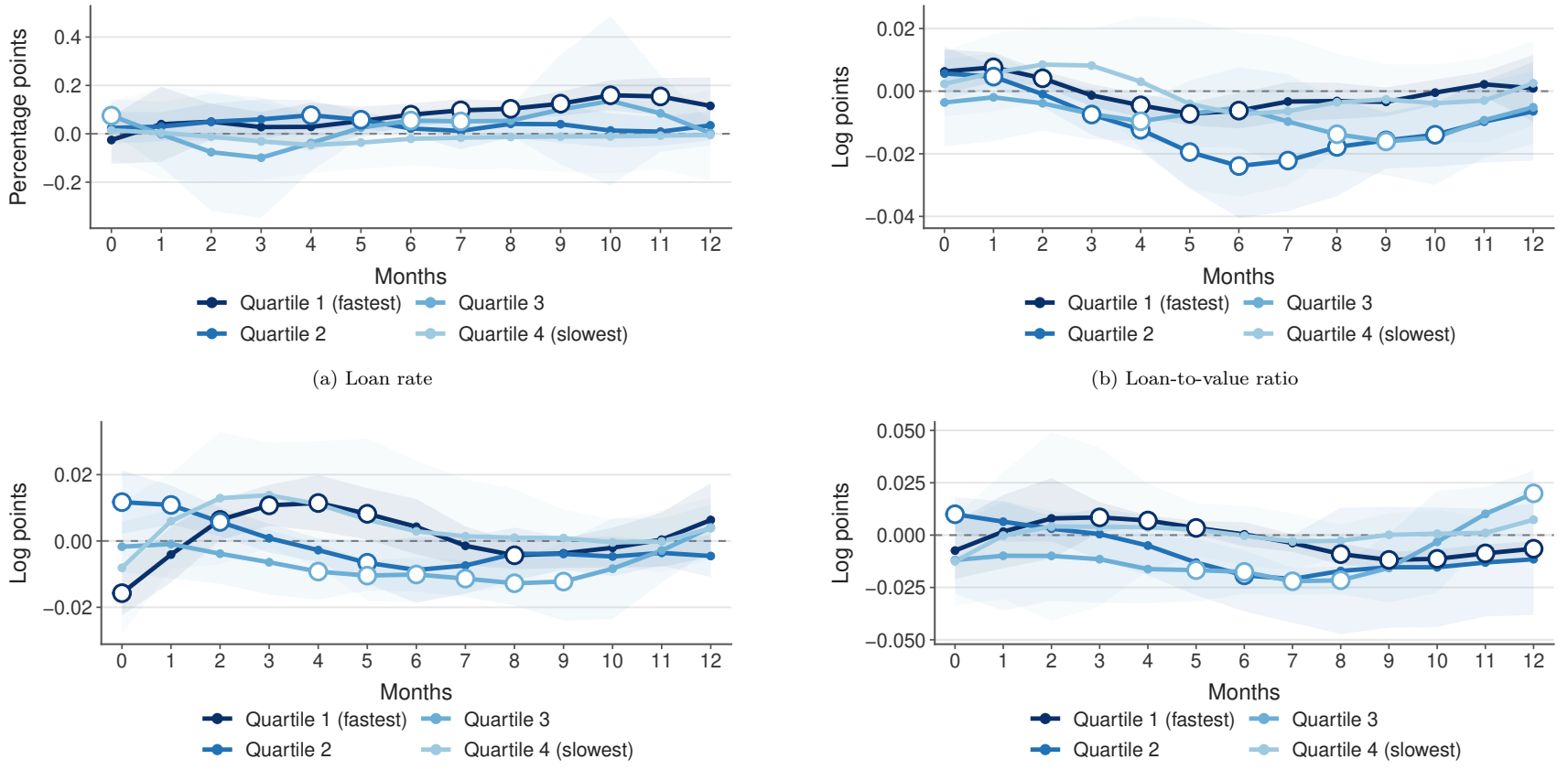
(a) Loan rate

(b) Loan-to-value ratio

(c) Maturity

(d) Loan amount

Figure 6: Responses of credit terms at captive lenders and specialized lenders relative to banks, by speed of pool entry. Each panel plots the difference between the response of a credit term at captive lenders, or at specialized lenders, and its response at banks with loans of the same pool speed in the same country and origination month, h months after a normalized monetary-policy surprise. A loan enters fast if it reaches a securitization pool within 584 days of origination, the sample median, and slowly otherwise (Table 4). The estimates come from one regression that interacts the lender-type terms of equation (8) with pool speed and adds pool speed to the fixed effects. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero.



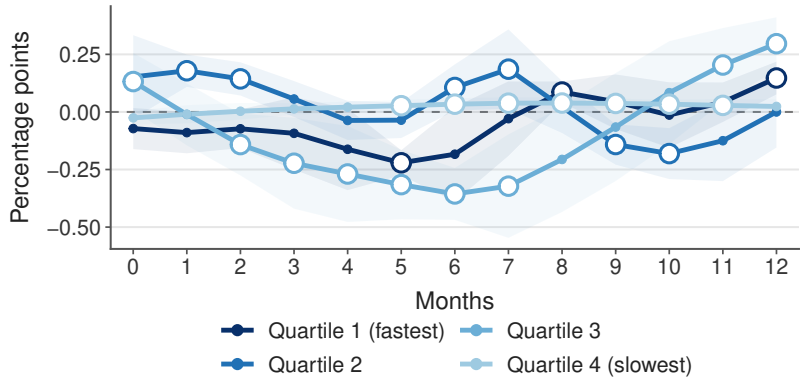
(a) Loan rate

(b) Loan-to-value ratio

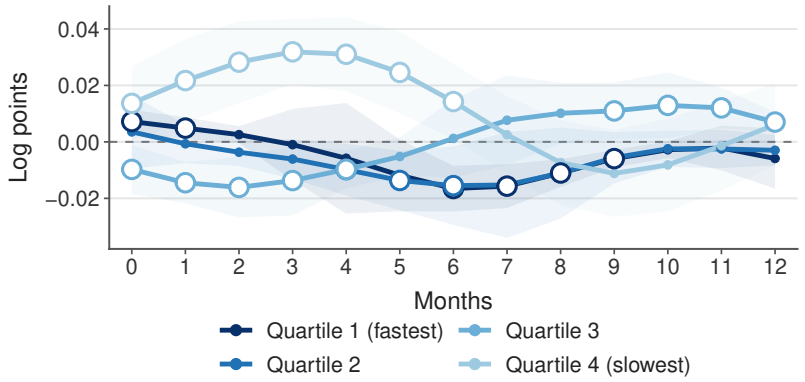
(c) Maturity

(d) Loan amount

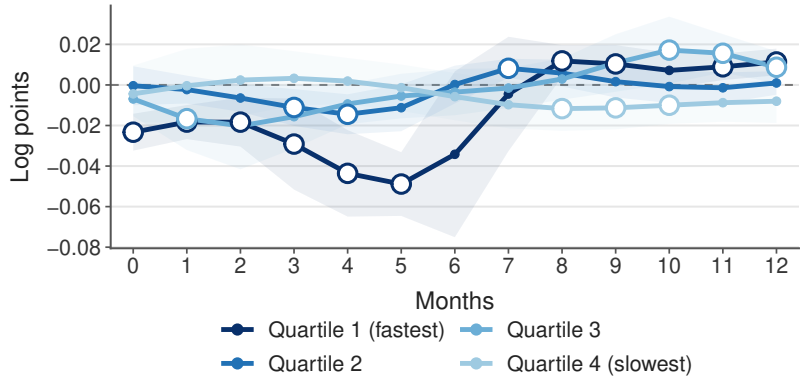
Figure 7: Responses of credit terms at captive lenders relative to banks, by quartile of time to pool entry. Each panel plots the difference between the response of a credit term at captive lenders and its response at banks among loans in the same quartile of days from origination to pool entry, in the same country and origination month, h months after a normalized monetary-policy surprise. Quartile 1 holds the loans that enter a pool within 311 days of origination and quartile 4 those that enter after more than 1,029 days (Table 4). The estimates come from one regression that interacts the lender-type terms of equation (8) with the pool-entry quartile and adds the quartile to the fixed effects; Figure 8 reports its specialized-lender terms. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. Shaded areas are pointwise 95 percent confidence intervals based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which the interval excludes zero.



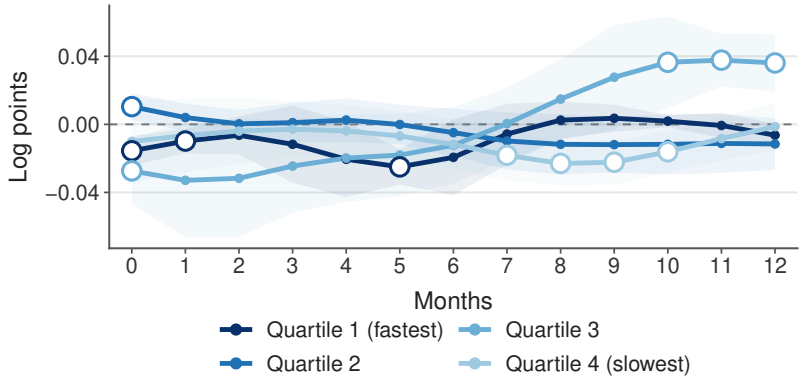
(a) Loan rate



(b) Loan-to-value ratio

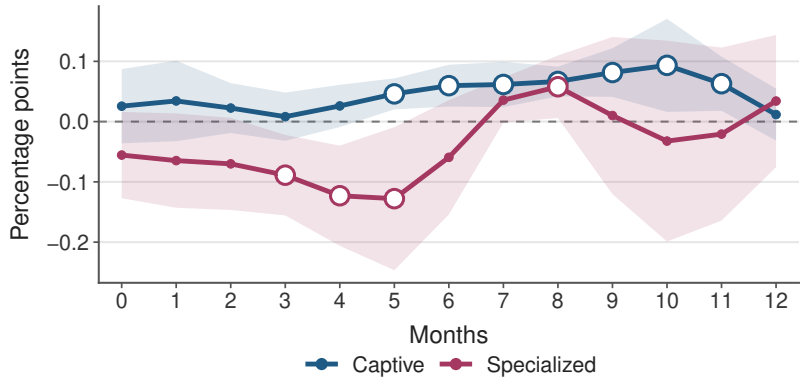


(c) Maturity

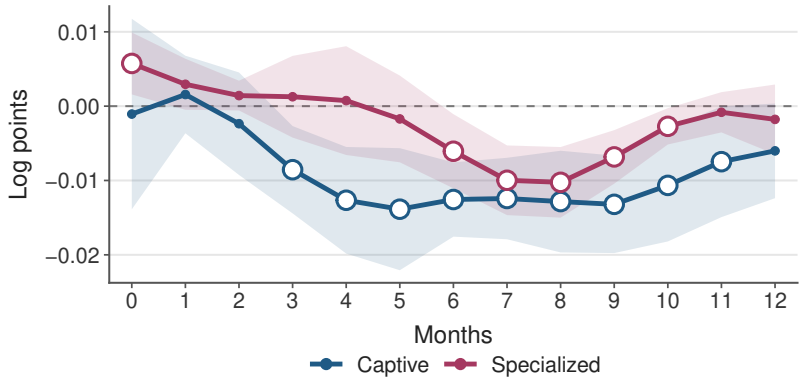


(d) Loan amount

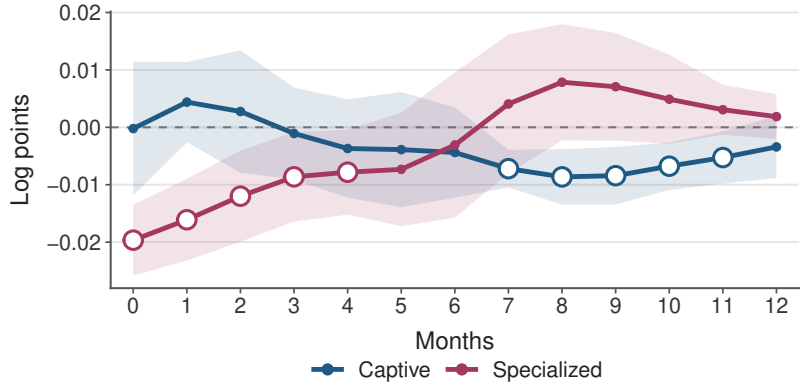
Figure 8: Responses of credit terms at specialized lenders relative to banks, by quartile of time to pool entry. The figure repeats Figure 7 for specialized lenders, from the same regression. Units, confidence intervals and open markers are as in Figure 7.



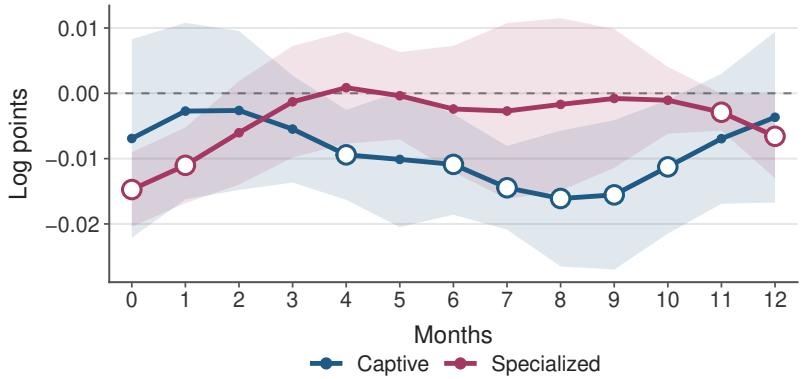
(a) Loan rate



(b) Loan-to-value ratio

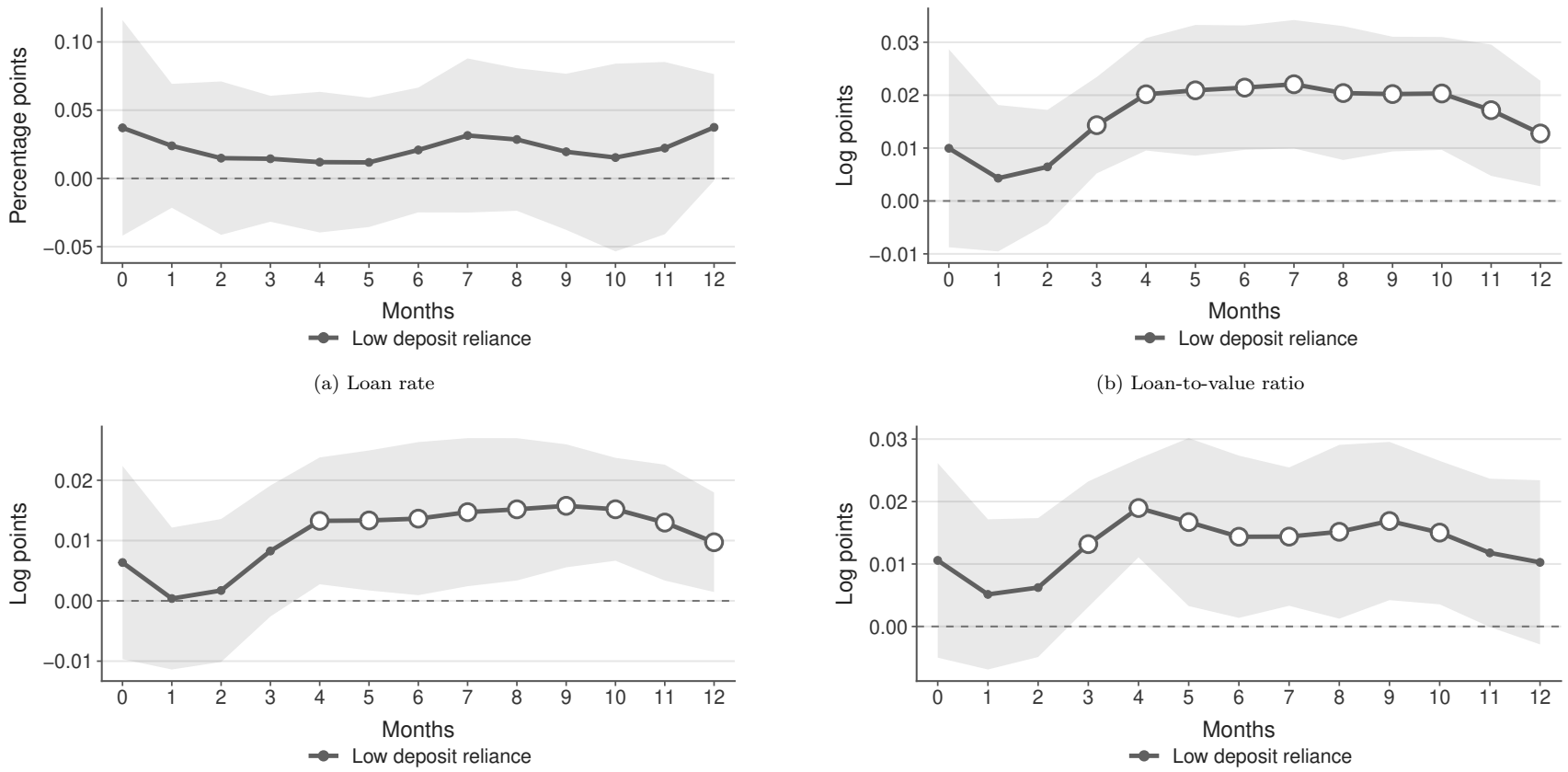


(c) Maturity



(d) Loan amount

Figure 9: Responses of credit terms at captive lenders and specialized lenders relative to banks, excluding Germany. The figure repeats Figure 1 on loans made in Spain, France, Italy and Portugal. Units, confidence intervals and open markers are as in Figure 1.



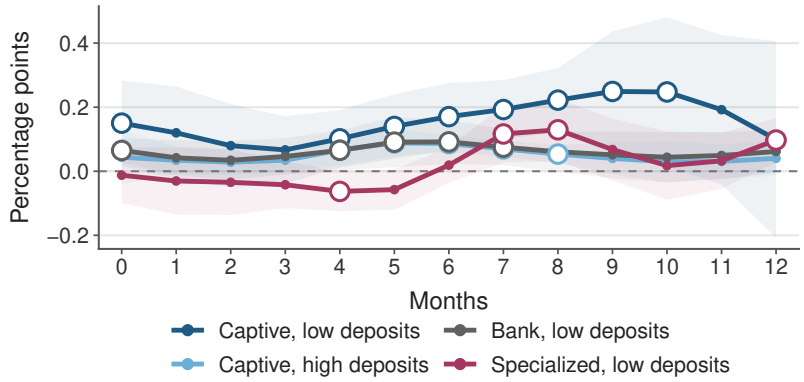
(a) Loan rate

(b) Loan-to-value ratio

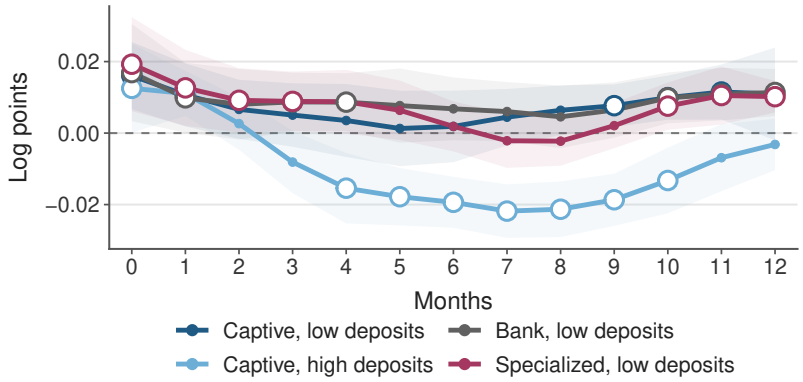
(c) Maturity

(d) Loan amount

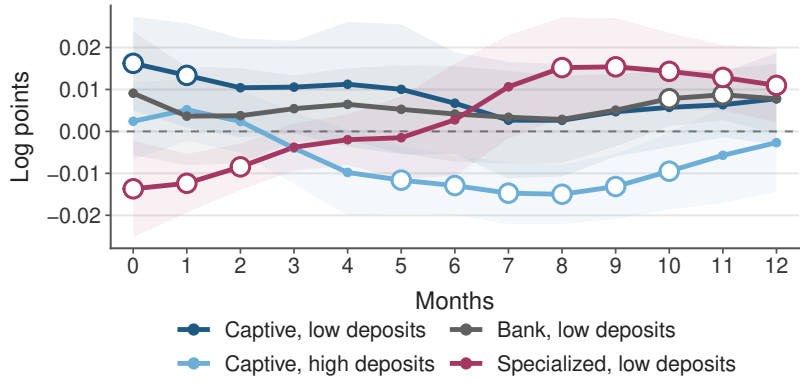
Figure 10: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance, excluding Germany. The figure repeats Figure 3 on loans made in Spain, France, Italy and Portugal. Units, confidence intervals and open markers are as in Figure 3.



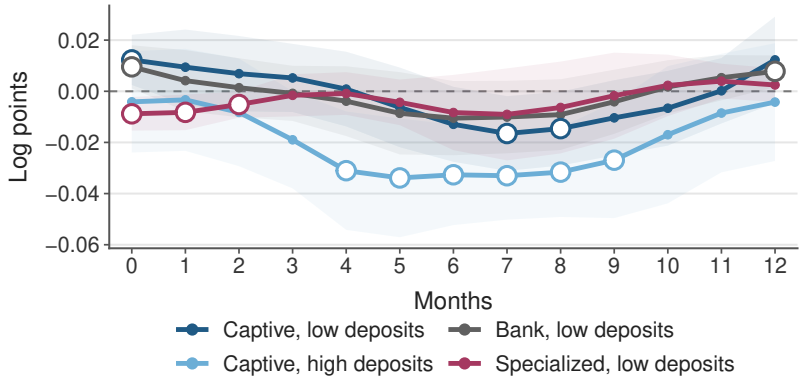
(a) Loan rate



(b) Loan-to-value ratio



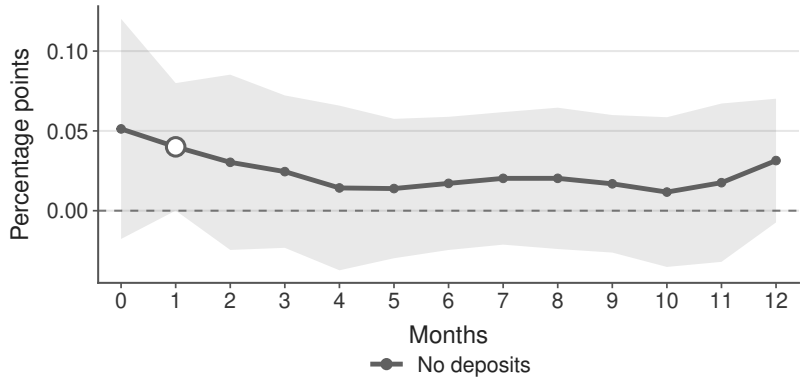
(c) Maturity



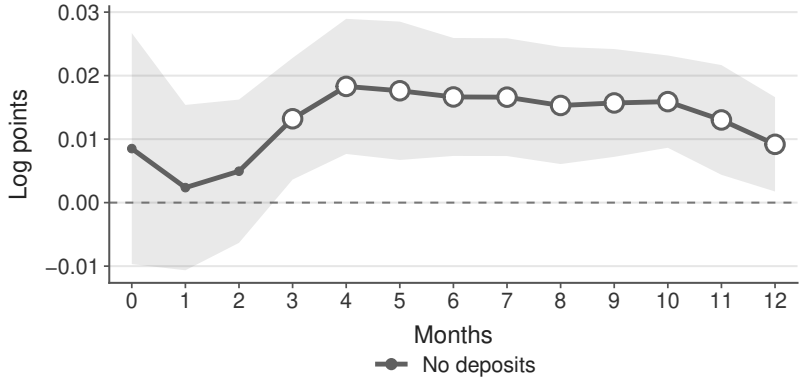
(d) Loan amount

Figure 11: Responses of credit terms by lender type and deposit reliance relative to banks with high deposit reliance, excluding Germany. The figure repeats Figure 4 on loans made in Spain, France, Italy and Portugal. Units, confidence intervals and open markers are as in Figure 4.

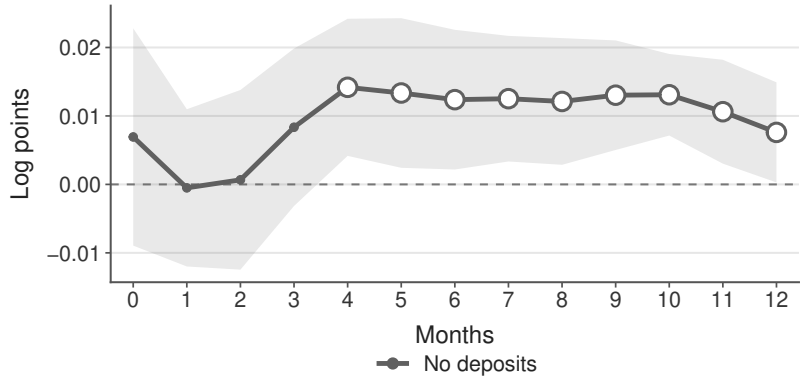
B.2 Funding and shock definitions



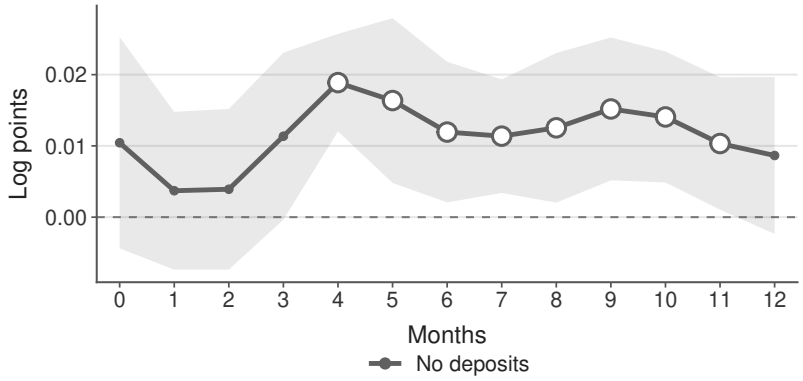
(a) Loan rate



(b) Loan-to-value ratio

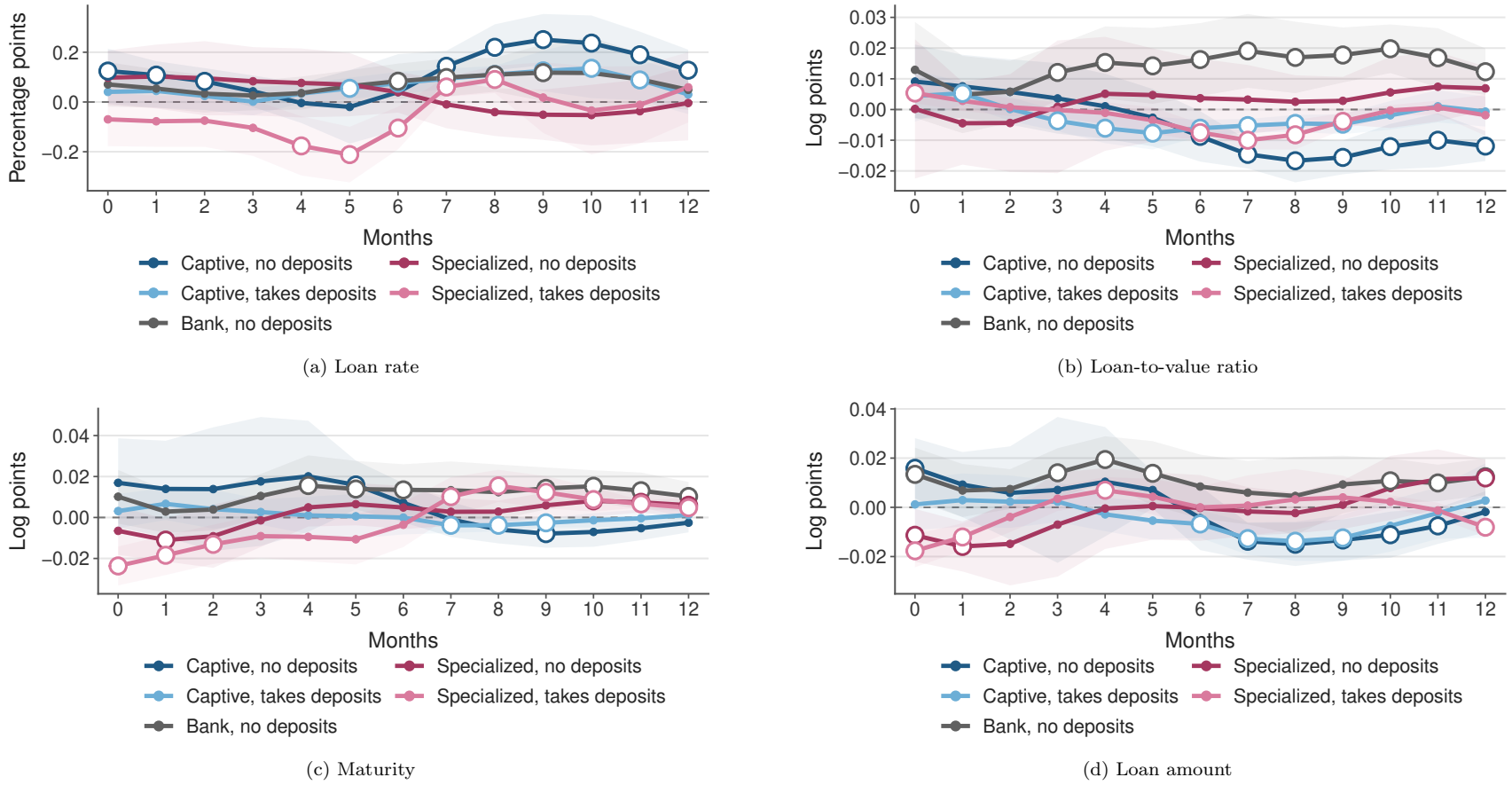


(c) Maturity



(d) Loan amount

Figure 12: Responses of credit terms at lenders that take no deposits relative to lenders that take deposits. The figure repeats Figure 3 with lenders grouped by whether they report positive deposits in any year from 2014 to 2024 (Table 5). Units, confidence intervals and open markers are as in Figure 3.



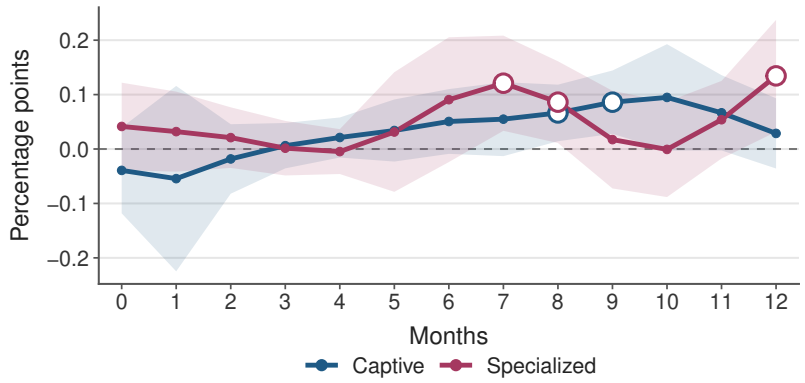
(a) Loan rate

(b) Loan-to-value ratio

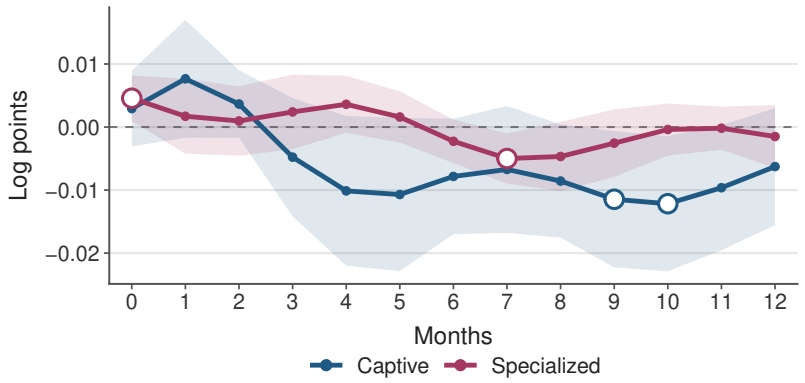
(c) Maturity

(d) Loan amount

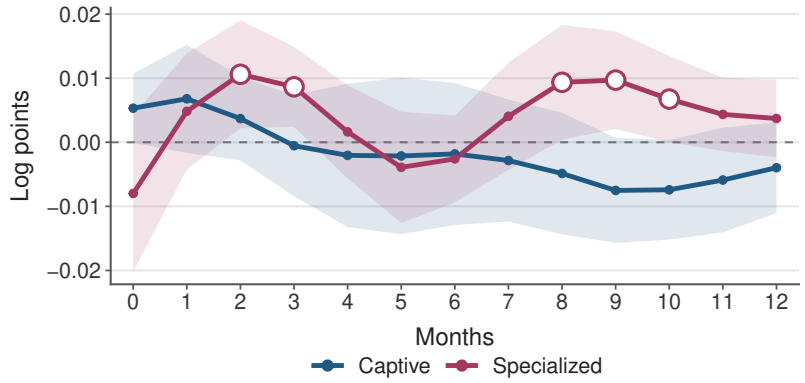
Figure 13: Responses of credit terms by lender type and deposit taking relative to banks that take deposits. The figure repeats Figure 4 with lenders grouped by whether they report positive deposits in any year from 2014 to 2024 (Table 5). One specialized lender takes deposits under this grouping, so the figure adds specialized lenders that take deposits. Units, confidence intervals and open markers are as in Figure 4.



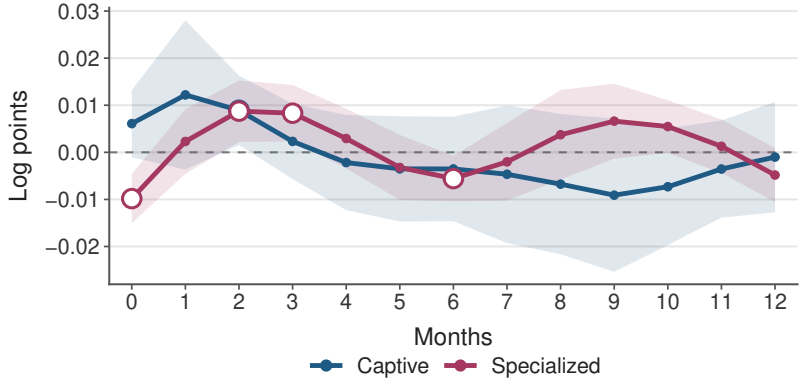
(a) Loan rate



(b) Loan-to-value ratio

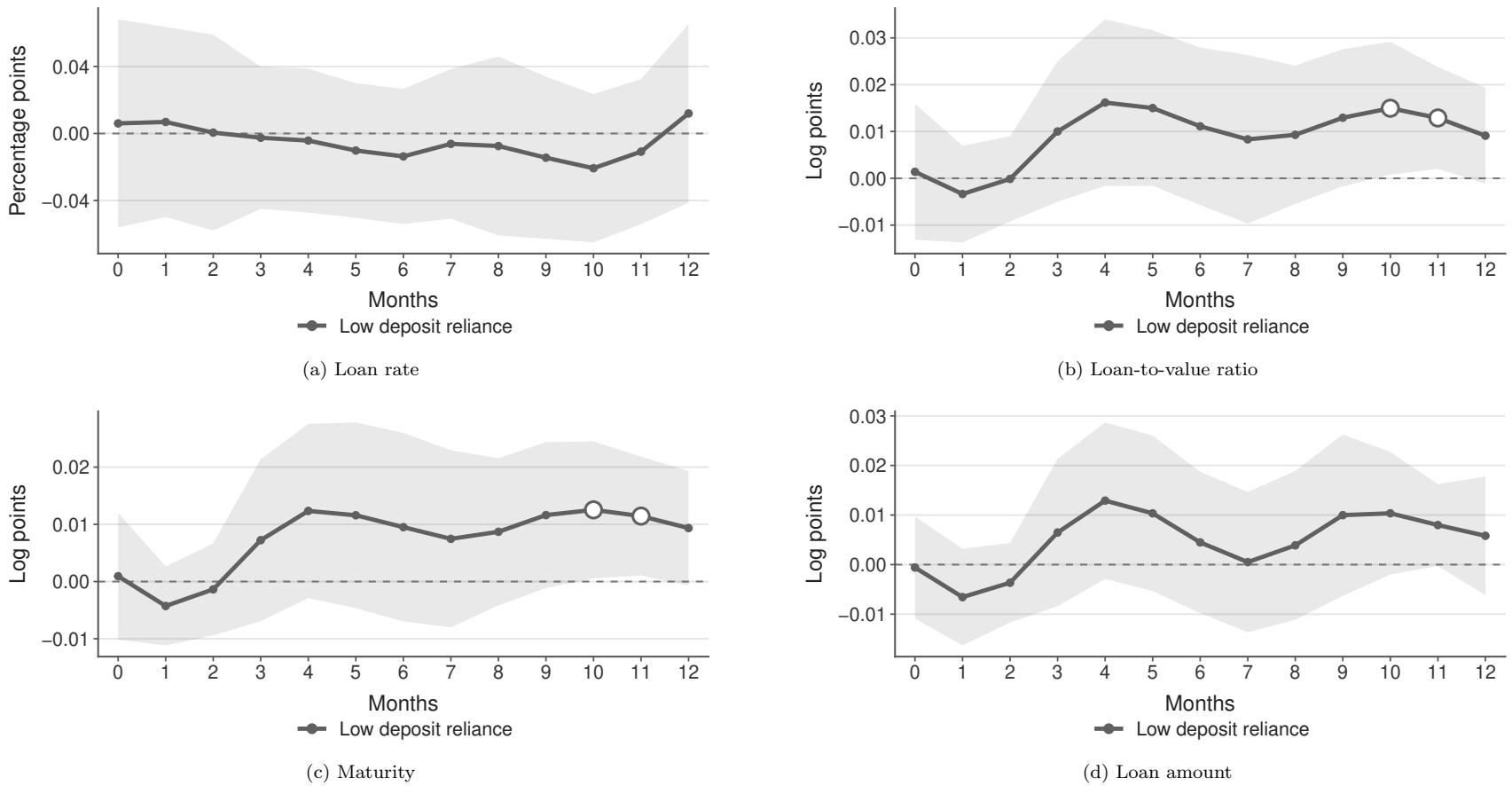


(c) Maturity



(d) Loan amount

Figure 14: Responses of credit terms at captive lenders and specialized lenders relative to banks, total monetary-policy surprise. The figure repeats Figure 1 with the surprise measured by the change in the one-week OIS rate around ECB Governing Council announcements, without separating the pure policy component from the information component, weighted across the month as in equation (7) and divided by 0.0785759. Units, confidence intervals and open markers are as in Figure 1.



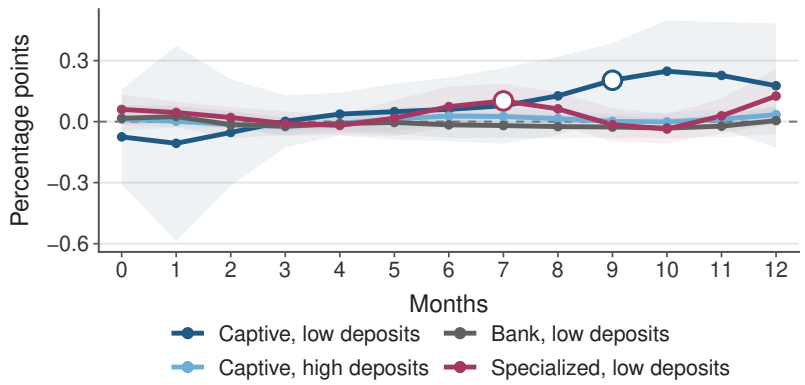
(a) Loan rate

(b) Loan-to-value ratio

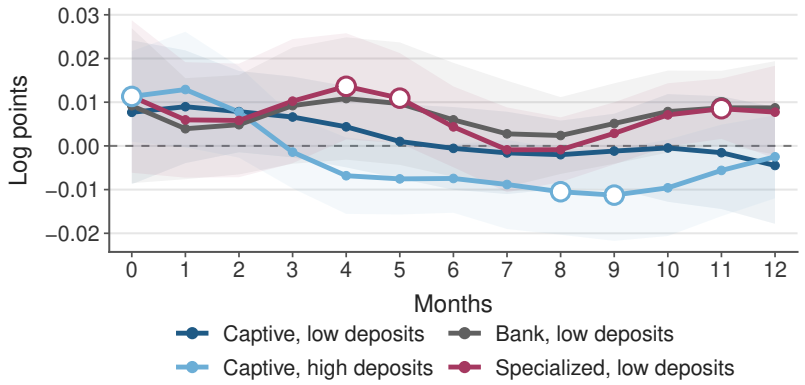
(c) Maturity

(d) Loan amount

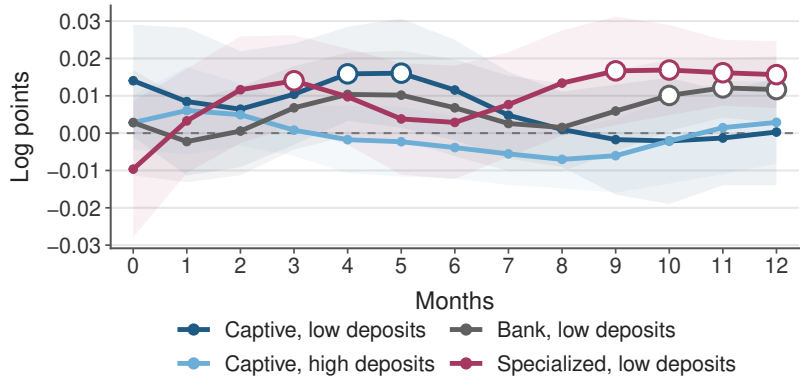
Figure 15: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance, total monetary-policy surprise. The figure repeats Figure 3 with the surprise of Figure 14. Units, confidence intervals and open markers are as in Figure 3.



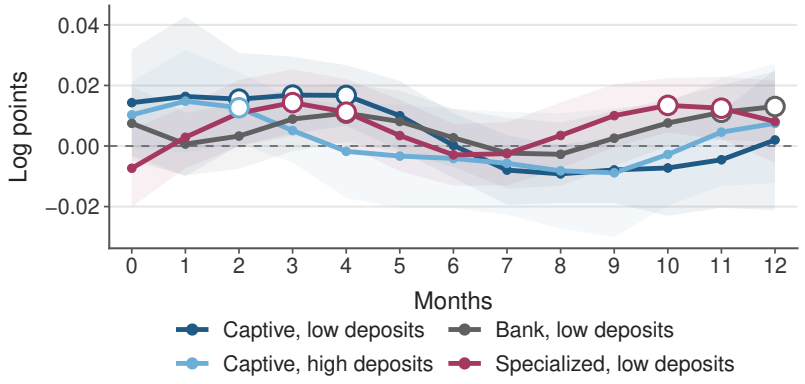
(a) Loan rate



(b) Loan-to-value ratio

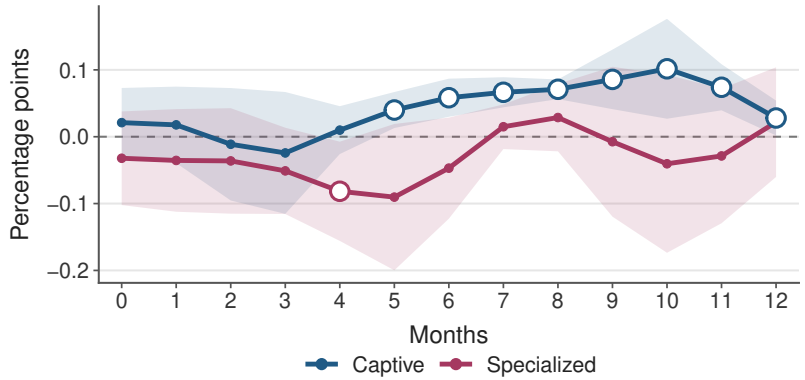


(c) Maturity

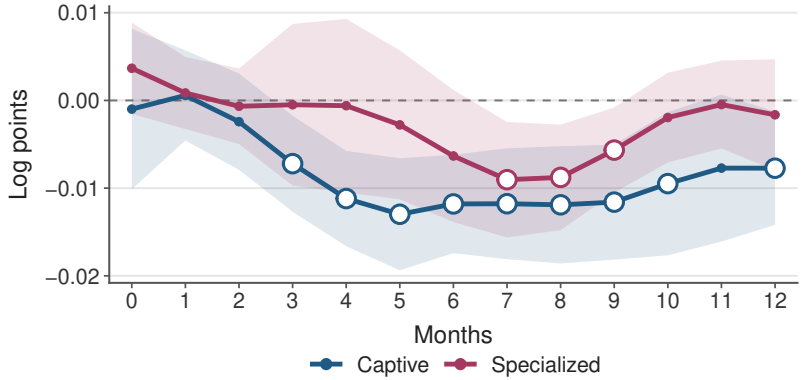


(d) Loan amount

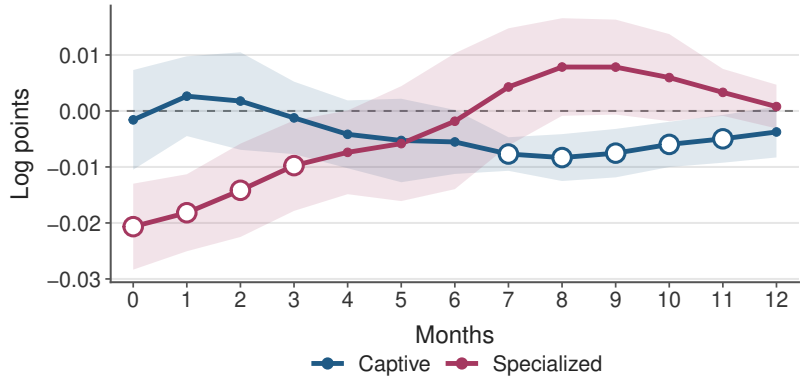
Figure 16: Responses of credit terms by lender type and deposit reliance relative to banks with high deposit reliance, total monetary-policy surprise. The figure repeats Figure 4 with the surprise of Figure 14. Units, confidence intervals and open markers are as in Figure 4.



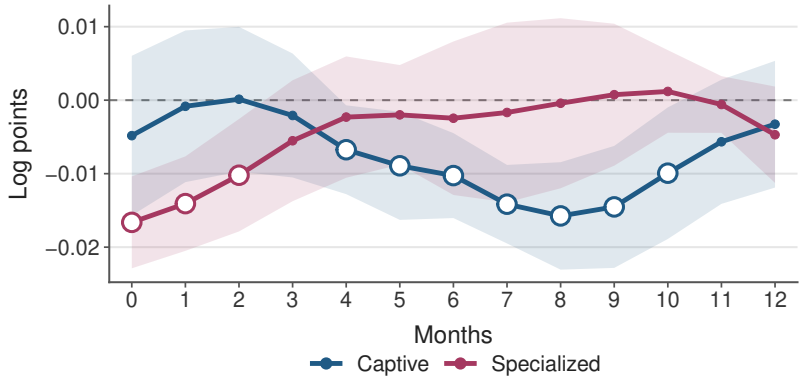
(a) Loan rate



(b) Loan-to-value ratio

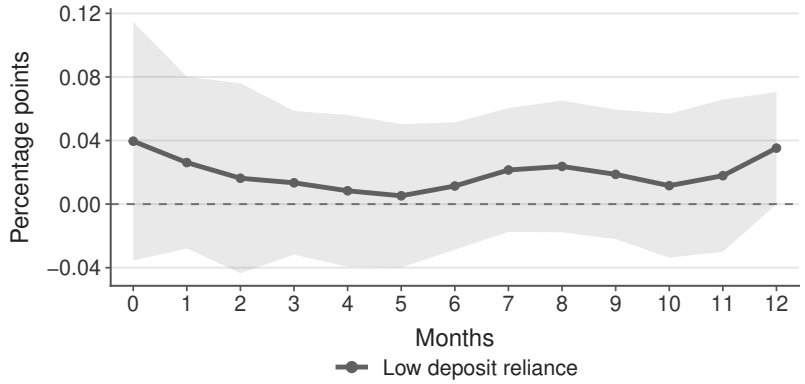


(c) Maturity

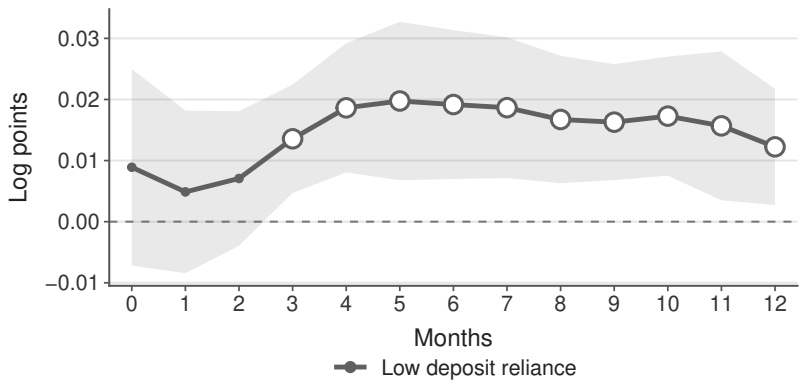


(d) Loan amount

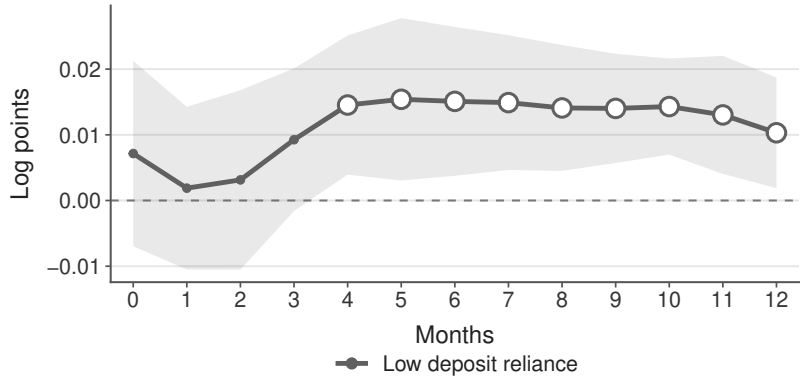
Figure 17: Responses of credit terms at captive lenders and specialized lenders relative to banks, announcement days weighted equally. The figure repeats Figure 1 with the pure monetary-policy surprises averaged over the calendar days of the month, so that each announcement receives the same weight whatever its date, and divided by 0.0358613. Units, confidence intervals and open markers are as in Figure 1.



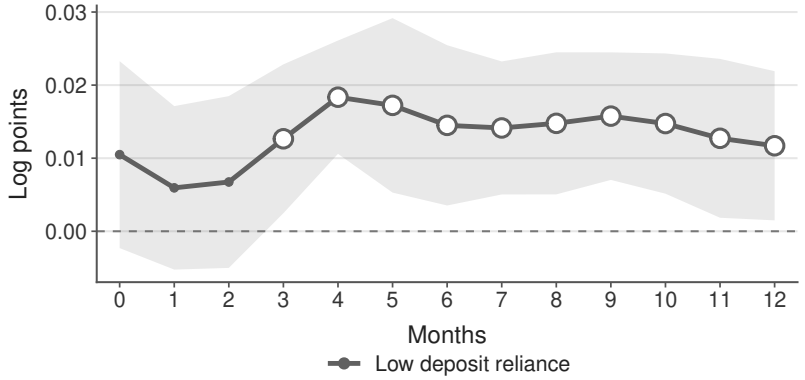
(a) Loan rate



(b) Loan-to-value ratio

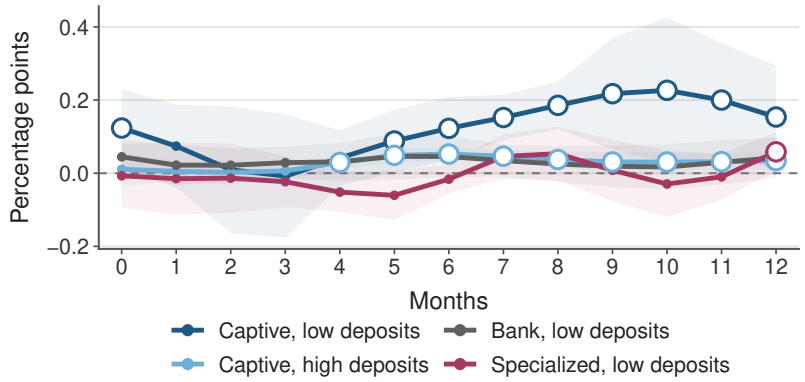


(c) Maturity

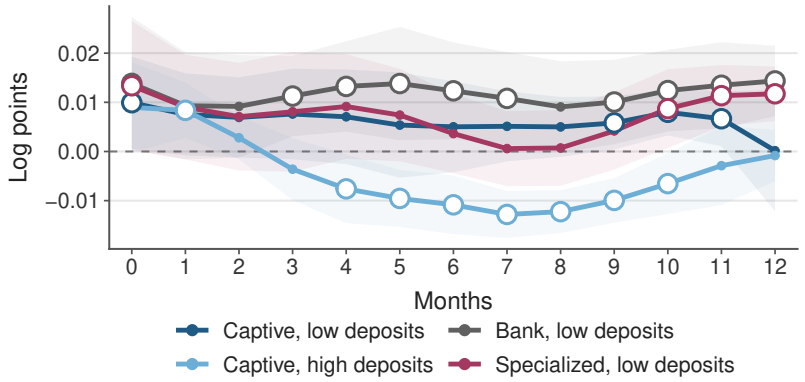


(d) Loan amount

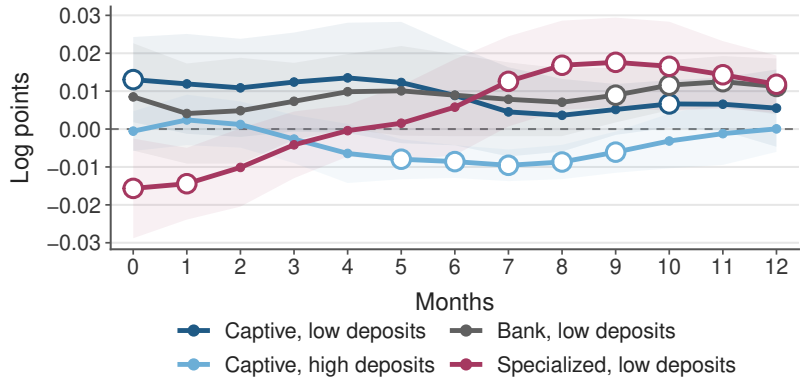
Figure 18: Responses of credit terms at lenders with low deposit reliance relative to lenders with high deposit reliance, announcement days weighted equally. The figure repeats Figure 3 with the surprise of Figure 17. Units, confidence intervals and open markers are as in Figure 3.



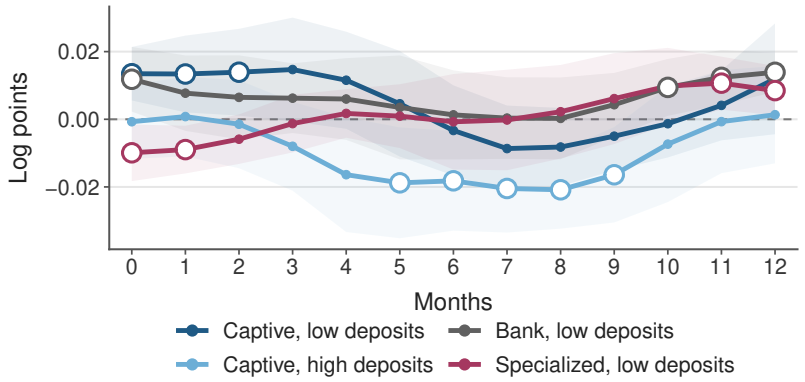
(a) Loan rate



(b) Loan-to-value ratio



(c) Maturity

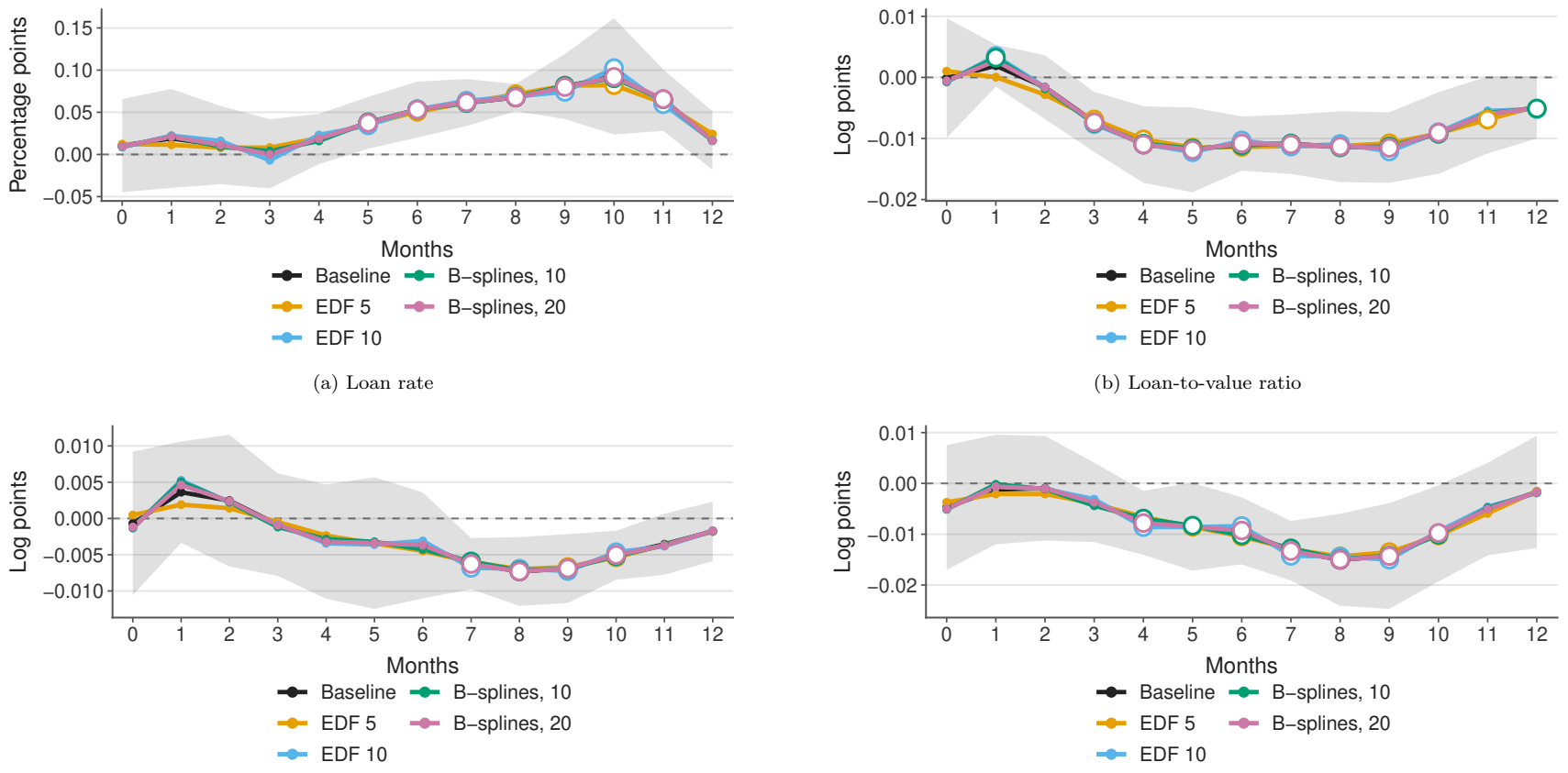


(d) Loan amount

Figure 19: Responses of credit terms by lender type and deposit reliance relative to banks with high deposit reliance, announcement days weighted equally. The figure repeats Figure 4 with the surprise of Figure 17. Units, confidence intervals and open markers are as in Figure 4.

B.3 Estimator settings

The baseline sets the smoothing penalty so that each response path has 7.5 effective degrees of freedom, on a basis of 15 cubic B-splines with unit-spaced knots over horizons 0 to 12. Figures 20 and 21 re-estimate Figure 1 with targets of 5 and 10 effective degrees of freedom per path, and with cubic B-spline bases of 10 and 20 functions at the baseline target.



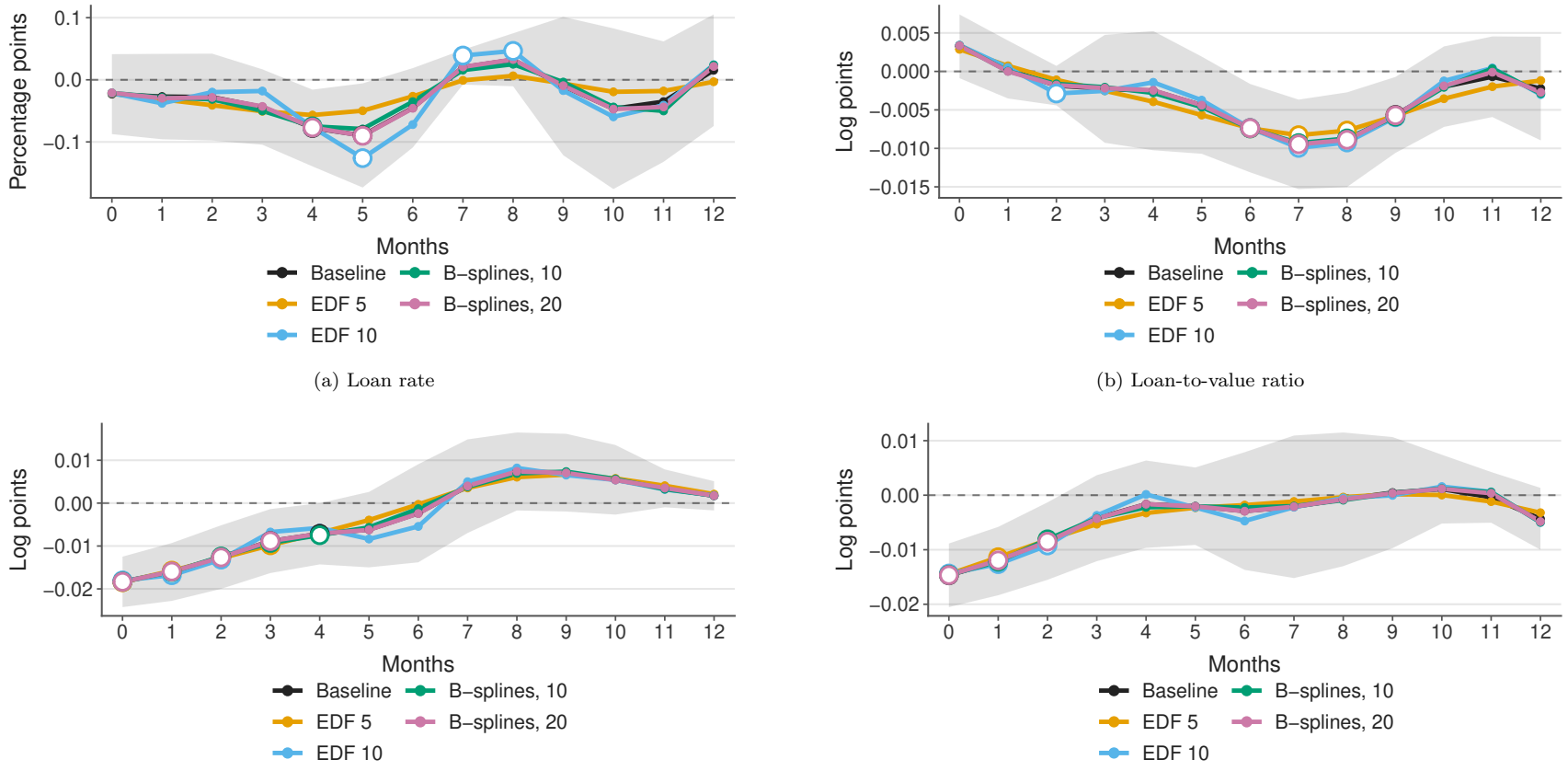
(a) Loan rate

(b) Loan-to-value ratio

(c) Maturity

(d) Loan amount

Figure 20: Responses of credit terms at captive lenders relative to banks under alternative smoothing settings. Each panel plots the captive–bank differential of Figure 1 with the baseline settings and with the four alternatives of Appendix B.3. Panel (a) is in percentage points; panels (b) to (d) are changes in logs. The shaded area is the pointwise 95 percent confidence interval for the baseline, based on Driscoll–Kraay standard errors by shock month with 12 lags, and open markers show horizons at which a series' own interval excludes zero.



(a) Loan rate

(b) Loan-to-value ratio

(c) Maturity

(d) Loan amount

Figure 21: Responses of credit terms at specialized lenders relative to banks under alternative smoothing settings. The figure repeats Figure 20 for the specialized–bank differential. Units, confidence intervals and open markers are as in Figure 20.